FOUNDATIONS OF

THEORETICAL PHYSICS AND QUANTUM CHEMISTRY

A TWO-PART COMPENDIUM OF FOUNDATIONAL FORMULAS

PART I:
QUANTUM GRAVITY & HOLOGRAPHY

PART II:
QUANTUM & COMPUTATIONAL CHEMISTRY

THIRTY-THREE SECTIONS, FROM THE COVARIANT PATH INTEGRAL
TO QUANTUM COMPUTING FOR CHEMISTRY

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

ULTRA

PHYSICAL CHEMISTRY

THE UNITY OF NATURE

$$Q = \sum_i g_i e^{-E_i/k_B T}$$

THERMODYNAMICS

$$dU = TdS - PdV + \mu dN$$

$$F = U - TS$$

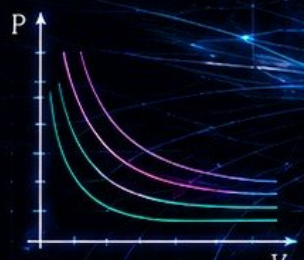
$$dG = -SdT + VdP + \mu dN$$

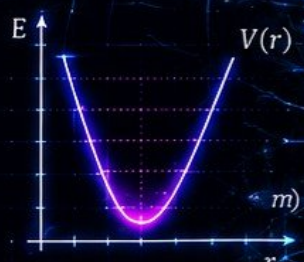
Maxwell Relations

STATISTICAL MECHANICS

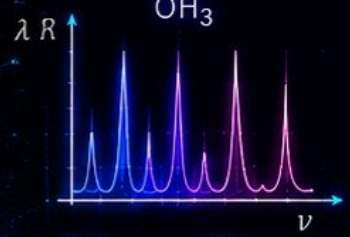
$$Z = \sum_i e^{-\beta E_i}$$

$$\langle A \rangle = \frac{1}{Z} \sum_i A_i e^{-\beta E_i}$$









GRAVITY · HOLOGRAPHY · QUANTUM · SPECTROSCOPY · MOLECULAR STRUCTURE · STATISTICS · KINETICS · ELECTROCHEMISTRY · QUANTUM COMPUTING

33 SECTIONS | 2 PARTS | 1 UNITY

ALIREZA MALEKIFAR

RIGOR • UNITY • INSIGHT • HARMONY

Foundations of Theoretical Physics and Quantum Chemistry

A Two-Part Compendium of Foundational Formulas

Part I: Quantum Gravity & Holography — Part II: Quantum & Computational Chemistry

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Thirty-Three Sections, from the Covariant Path Integral
to Quantum Computing for Chemistry

July 2026

Reference compendium; typeset for internal consistency of notation and cross-checked at a small number of specific points noted in footnotes.

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Intended Audience

Postdoctoral researchers in the quantum sciences

This compendium collects and annotates established results from the published literature; the selection, organization, cross-referencing, and annotations throughout are the compiler's own work.

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Preface: Scope, Conventions, and a Meta-Theoretical Caveat

This compendium collects foundational expressions from twenty distinct theoretical programs concerned with the quantization of gravity and its holographic reformulations: the covariant gravitational path integral, classical general relativity and black hole thermodynamics, higher-curvature and Lovelock gravity, Jackiw–Teitelboim (JT) dilaton gravity, Chern–Simons theory, the Plebanski–Holst formulation underlying loop quantum gravity, canonical loop quantum gravity (LQG) and loop quantum cosmology (LQC), spin foam models, group field theory (GFT), asymptotic safety, causal set theory, simplicial quantum gravity (Regge calculus and dynamical triangulations), random tensor and matrix models, the Sachdev–Ye–Kitaev (SYK) model, and the holographic dictionary (dS/CFT, AdS/CFT, entanglement entropy, islands, and complexity).

A caveat that bears emphasis. These are largely independent – and in several respects mutually incompatible or as-yet-unreconciled – research programs, pursued with different foundational commitments (e.g., background independence versus a fixed asymptotic boundary; fundamental discreteness of geometry versus a continuum path integral). Their juxtaposition here serves reference and pedagogical convenience; it should not be read as implying that they are complementary pieces of one settled, unified theory of quantum gravity. Where two sections concern structurally related objects (e.g., JT gravity in Section 4 and its matrix-model dual in Section 15, or LQG in Sections 6–8 and its covariant spin-foam completion in Section 9), a cross-reference is given.

Conventions. Natural units $c = 1$ are used throughout; G and $\hbar$ are retained explicitly wherever the source convention does so (some LQG operator equations below are implicitly in units $\hbar = 1$, following that literature's own convention). A mostly-plus metric signature $(-,+,+,+)$ is used wherever a signature is implicit. Two path-integral weight conventions coexist in this compendium, each inherited from the standard practice of the corresponding subfield: a *Lorentzian* weight e^{iS} (the covariant path integral, Chern–Simons/BF/Plebanski/Holst theory, canonical LQG) and a *Euclidean* weight e^{-S} (matrix and tensor models, causal sets, dynamical triangulations, and the disk-level JT partition function). This is a substantive feature of how these formulas are actually used in the literature, not a typographical inconsistency, and it is preserved here rather than artificially harmonized.

A small number of equations below carry a footnote flagging a genuine subtlety – a quantum level shift, a convention-dependent normalization, or a point of nonstandard notation – rather than silently amending the expression as supplied. Every equation is reproduced exactly as given; captions that state a standard result for comparison are clearly marked as such, and the handful of footnoted cross-checks were verified against the primary literature rather than against recollection alone.

Part I

Part I — Quantum Gravity and Holography

1. Covariant Path-Integral Quantization of Gravity

The most direct extrapolation of the Feynman path integral to gravity sums over off-shell metrics (and matter configurations) weighted by e^{iS}. It is purely formal at the non-perturbative level – the measure $\mathcal{D}g_{\mu\nu}$ has no known rigorous, diffeomorphism-invariant definition in $d = 4$ – but it organizes perturbation theory around a fixed background and motivates virtually every discretized or canonical scheme discussed in later sections.

$$\mathcal{Z}_{\text{quantum gravity}} = \int \mathcal{D}g_{\mu\nu}\,\mathcal{D}\phi\,\mathcal{D}A_\mu\,\mathcal{D}\psi\,\mathcal{D}\bar{\psi}\; e^{iS[g,\phi,A,\psi]},$$
$$S = \int d^4x\,\sqrt{-g}\left[\frac{1}{16\pi G}(R - 2\Lambda) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\,i\gamma^\mu D_\mu\psi + \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(\phi)\right] \tag{1.1}$$

The formal gravitational path integral for Einstein gravity with cosmological constant Λ, minimally coupled to a $U(1)$ gauge field (Maxwell term), a Dirac fermion (via the gauge- and spin-covariant derivative D_μ), and a real scalar with potential $V(\phi)$ – a toy stand-in for coupling gravity to Standard-Model-type matter.

$$\langle 0|\,T\{\phi(x_1)\cdots\phi(x_n)\}\,|0\rangle = \frac{1}{\mathcal{Z}}\int \mathcal{D}\phi\;\phi(x_1)\cdots\phi(x_n)\,e^{iS[\phi]} \tag{1.2}$$

Path-integral representation of a vacuum time-ordered n-point function, in the Lorentzian ("in–in") formulation of quantum field theory; the gravitational path integral above is the same construction applied to $g_{\mu\nu}$ in place of ϕ.

$$G_F(x,y) = \int \frac{d^4p}{(2\pi)^4}\,\frac{i}{p^2 - m^2 + i\epsilon}\,e^{ip\cdot(x-y)} \tag{1.3}$$

The momentum-space Feynman propagator for a free scalar of mass m; the $i\epsilon$ prescription selects the time-ordered (Feynman) contour and is what renders the p^0-integral convergent.

$$\hat{\mathcal{H}}\,\Psi[g_{\mu\nu}] = 0 \tag{1.4}$$

The Wheeler–DeWitt equation: the canonical (Hamiltonian) counterpart of the covariant path integral above, obtained by promoting the ADM Hamiltonian constraint of general relativity to an operator $\hat{\mathcal{H}}$ annihilating the wavefunctional of the universe $\Psi[g_{\mu\nu}]$. The absence of an explicit time parameter in this equation – time enters only implicitly, through the dynamical variables themselves – is the origin of the "problem of time" in canonical quantum gravity.

$$\mathcal{Z} \approx \sum_{\text{cl}} e^{iS_{\text{cl}}/\hbar} \left[\det(-S''_{\text{cl}})\right]^{-1/2} \big(1 + O(\hbar)\big) \tag{1.5}$$

The semiclassical (stationary-phase) approximation to the path integral: dominated by classical solutions S_{cl} of the equations of motion, with the one-loop determinant of second variations giving the leading quantum correction; applied to Eq. 1.1, this is the route by which the path integral reproduces the classical field equations of Section 2 as $\hbar \to 0$, and by which black hole free energies are computed from the on-shell Euclidean action.

$$g_{\mu\nu}dx^\mu dx^\nu = -N^2dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt) \tag{1.6}$$

The ADM (3+1) decomposition: foliating spacetime into spatial slices of induced metric h_{ij}, with lapse N and shift N^i encoding how successive slices are stacked; this is the starting point for *every* canonical formulation in this compendium (the Wheeler–DeWitt equation above, and the Ashtekar–Barbero variables of Section 6) as opposed to the covariant path-integral treatment of Eq. 1.1.

$$\frac{d\theta}{d\tau} = -\frac{\theta^2}{d-2} - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu \tag{1.7}$$

The Raychaudhuri equation: the exact evolution of the expansion θ of a congruence of timelike (or null) geodesics with tangent u^μ, in terms of shear $\sigma_{\mu\nu}$, rotation $\omega_{\mu\nu}$, and the Ricci curvature sourced (via Eq. 2.3) by matter. For a hypersurface-orthogonal congruence ($\omega = 0$) satisfying the null energy condition, every term on the right is non-positive, forcing $\theta \to -\infty$ in finite affine parameter – the purely geometric focusing argument underlying the Penrose–Hawking singularity theorems, which show singularities are a generic, not fine-tuned, feature of gravitational collapse.

2. Classical General Relativity: The Einstein–Hilbert Action and Black Hole Thermodynamics

Every covariant approach to quantum gravity must reproduce, in the appropriate limit, the classical dynamics obtained by extremizing the Einstein–Hilbert action. This section fixes that classical baseline, together with the thermodynamic quantities that later sections (JT gravity, holographic entanglement entropy, islands) generalize or re-derive from a more microscopic starting point.

$$S_{\text{EH}} = \frac{1}{16\pi G}\int d^4x\,\sqrt{-g}\,(R - 2\Lambda) \tag{2.1}$$

The Einstein–Hilbert action with cosmological constant Λ.

$$\delta S_{\text{EH}} = \frac{1}{16\pi G}\int d^4x\,\sqrt{-g}\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu}\right)\delta g^{\mu\nu} \tag{2.2}$$

Metric variation of Eq. (2.1); the boundary term produced by integrating by parts is exactly canceled by the Gibbons–Hawking–York term of Eq. (3.4) and is omitted here.

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu} \tag{2.3}$$

The Einstein field equations, obtained by demanding $\delta(S_{\mathrm{EH}} + S_{\mathrm{matter}}) = 0$ against Eq. (2.2) and the definition of $T_{\mu\nu}$ in Eq. (2.4).

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta S_{\mathrm{matter}}}{\delta g^{\mu\nu}} \tag{2.4}$$

The Hilbert (metric) definition of the stress-energy tensor, sourcing Eq. (2.3).

$$ds^2 = -\left(1 - \frac{2GM}{r} + \frac{r^2}{\ell^2}\right) dt^2 + \left(1 - \frac{2GM}{r} + \frac{r^2}{\ell^2}\right)^{-1} dr^2 + r^2 d\Omega^2 \tag{2.5}$$

The static, spherically symmetric, asymptotically anti-de Sitter black hole solution (Schwarzschild–AdS$_4$) of Eq. (2.3) in vacuum, with AdS radius ℓ (equivalently $\Lambda = -3/\ell^2$) and horizon function $f(r) = 1 - 2GM/r + r^2/\ell^2$.

$$T_H = \frac{\hbar \kappa}{2\pi}, \qquad \kappa = \frac{r_+ - r_-}{2 r_+^2} \tag{2.6}$$

The Hawking temperature in terms of the surface gravity κ at the outer horizon.[1]

$$S_{\mathrm{BH}} = \frac{A}{4G\hbar}, \qquad A = 4\pi r_+^2 \tag{2.7}$$

The Bekenstein–Hawking entropy, proportional to the horizon area A in Planck units – the benchmark that any microscopic theory of quantum gravity is required to reproduce (Section 19 revisits this via holographic entanglement entropy).

$$dM = T\, dS + \Omega\, dJ + \Phi\, dQ, \qquad M = 2TS + 2\Omega J + \Phi Q \tag{2.8}$$

The first law of black hole mechanics (Bardeen–Carter–Hawking) and the Smarr formula: the differential and integrated statements, respectively, of black hole thermodynamics for a stationary Kerr–Newman-type horizon – the direct analogue of $dU = TdS - pdV$, with the Smarr relation following from the homogeneity (scaling) properties of $M(S, J, Q)$.

[1] The formula given for κ in terms of an inner and outer horizon, r_- and r_+, is the standard result for a spacetime with *two* horizons (e.g. Reissner–Nordström or its AdS generalization). It does not apply as written to the single-horizon Schwarzschild–AdS metric directly above unless charge or angular momentum is reinstated; for that metric $\kappa = \frac{1}{2} f'(r_+)$ directly, which is in fact the general definition from which the two-horizon formula is itself derived.

$$M^2 = M_{\text{irr}}^2 + \frac{J^2}{4M_{\text{irr}}^2}, \qquad M_{\text{irr}} = \sqrt{\frac{A}{16\pi G^2}} \tag{2.9}$$

The Christodoulou irreducible mass: decomposes a Kerr black hole's mass into a horizon-area ("irreducible", non-extractable) piece and a rotational piece; the Penrose process can extract energy and reduce M toward M_{irr}, but never below it, since A (and hence M_{irr}) cannot decrease – an early, purely classical hint of the second law that Eq. 2.7 later promotes to a genuine thermodynamic statement.

$$S(b{:}c) + S(b{:}c') \leq 2\,S(b) \tag{2.10}$$

The entanglement-monogamy bound underlying the black hole firewall (AMPS) paradox: if an old black hole's early radiation b is to purify a late Hawking mode c (as unitarity, via the Page curve of Eq. 19.5, requires) *and* c is to be maximally entangled with its interior partner mode c' (as a smooth, "no-drama" horizon, i.e. the equivalence principle, requires), monogamy of entanglement – this inequality, itself a consequence of strong subadditivity – is violated; one of purity, effective-field-theory validity across the horizon, or no drama must fail for a sufficiently old black hole, with the island formula of Eq. 19.4 now understood as the resolution.

3. Higher-Curvature and Lovelock Extensions

Higher-derivative curvature invariants arise both as the UV counterterms required to renormalize semiclassical gravity and as the building blocks of the Lovelock family, the unique class of purely metric theories in $D > 4$ whose field equations remain second order in derivatives.

$$\mathcal{L}_{\text{higher curvature}} = \frac{1}{16\pi G}\sqrt{-g}\left[R + \alpha R^2 + \beta R_{\mu\nu}R^{\mu\nu} + \gamma R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right] \tag{3.1}$$

The general quadratic-curvature gravitational Lagrangian density; α, β, γ are Wilson coefficients of the dimension-four higher-derivative operators (this same truncation reappears, with a running-coupling interpretation, in Section 11 on asymptotic safety).

$$\mathcal{L}_{\text{Lovelock}} = \sum_{k=0}^{[d/2]} \alpha_k\, \mathcal{L}^{(k)} \tag{3.2}$$

The Lovelock Lagrangian in d spacetime dimensions: a sum of dimensionally continued Euler densities $\mathcal{L}^{(k)}$, each multiplying an independent coupling α_k ($k = 0$: cosmological constant; $k = 1$: Einstein–Hilbert; $k = 2$: Gauss–Bonnet; etc.), with only terms satisfying $2k \leq d$ contributing to the dynamics.

$$\mathcal{L}^{(k)} = \frac{1}{2^k}\,\delta^{\mu_1\cdots\mu_{2k}}_{\nu_1\cdots\nu_{2k}}\, R^{\nu_1\nu_2}_{\ \ \mu_1\mu_2}\cdots R^{\nu_{2k-1}\nu_{2k}}_{\ \ \mu_{2k-1}\mu_{2k}} \tag{3.3}$$

The k-th Lovelock density, built from k factors of the Riemann tensor fully antisymmetrized against a generalized (rank-$2k$) Kronecker delta.

$$S_{\text{Gibbons–Hawking}} = \frac{1}{8\pi G}\int_{\partial M} d^{d-1}x\,\sqrt{h}\,K \tag{3.4}$$

The Gibbons–Hawking–York boundary term: K is the trace of the extrinsic curvature of ∂M and h its induced metric. Required for a well-posed Dirichlet variational principle (it cancels the boundary term dropped in Eq. (2.2)); its on-shell value dominates holographic and Euclidean black-hole free energies alike.

$$S_{\text{Wald}} = -2\pi\oint_{\mathcal{H}} d^{d-2}x\,\sqrt{h}\,\frac{\partial\mathcal{L}}{\partial R_{\mu\nu\rho\sigma}}\,\epsilon_{\mu\nu}\epsilon_{\rho\sigma} \tag{3.5}$$

Wald's entropy formula: the generalization of the Bekenstein–Hawking area law (Eq. 2.7) to an arbitrary diffeomorphism-invariant Lagrangian $\mathcal{L}$, obtained as the functional derivative of $\mathcal{L}$ with respect to the Riemann tensor, contracted with the horizon binormal $\epsilon_{\mu\nu}$ and integrated over the bifurcation surface $\mathcal{H}$; for $\mathcal{L} = R/16\pi G$ it reduces exactly to $A/4G$, and for the higher-curvature Lagrangians of Section 3 it is generically *not* simply proportional to the area.

$$f(R)\text{ gravity:}\qquad f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + \big(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu\big)f'(R) = 8\pi G\,T_{\mu\nu} \tag{3.6}$$

The field equations of $f(R)$ gravity, the simplest higher-curvature theory (a special case of Eq. 3.1 with $\beta = \gamma = 0$): note the extra derivative terms relative to Eq. 2.3, which make $f(R)$ gravity a fourth-order theory generically propagating an additional scalar degree of freedom – the reason such theories require the Wald entropy formula above rather than the bare area law.

$$m \le \sqrt{2}\,G^{1/2}\,q \tag{3.7}$$

The Weak Gravity Conjecture (in natural units, for a $U(1)$ gauge theory coupled to gravity): the theory must contain a state whose charge-to-mass ratio exceeds that of an extremal black hole, i.e. gravity must be the weakest force. First proposed as a consistency requirement of quantum gravity (ensuring extremal black holes can discharge and decay rather than piling up as stable remnants), it is the best-studied example of the broader "swampland" program of conjectured constraints separating effective field theories that do, and do not, admit a UV completion in quantum gravity.

$$S(A) \le \frac{\text{Area}(A)}{4G} \tag{3.8}$$

Bousso's covariant entropy bound: the entropy passing through any light-sheet of a surface A (a null hypersurface generated by non-expanding light rays) is bounded by the surface's area in Planck units – a generalization of the Bekenstein–Hawking area law of Eq. 2.7 to

arbitrary surfaces, not just black hole horizons, and one of the earliest purely geometric hints that a consistent quantum gravity must be, in some precise sense, lower-dimensional than a naive field-theoretic degree-of-freedom count would suggest – the "holographic principle" that Sections 17–19 realize concretely.

4. Two-Dimensional Dilaton Gravity: Jackiw–Teitelboim (JT) Theory

JT gravity is the simplest solvable model of quantum gravity with a genuine holographic interpretation: a two-dimensional dilaton-gravity theory whose boundary dynamics (the Schwarzian mode) is dual to the near-conformal limit of the SYK model (Section 16), and whose exact non-perturbative quantization is captured by a matrix integral (Section 15).

$$S_{\mathrm{JT}} = \frac{\phi_0}{16\pi G}\left(\int d^2x\,\sqrt{-g}\,R + 2\int_\partial K\right) + \frac{1}{16\pi G}\int d^2x\,\sqrt{-g}\,\phi\,(R+2) \tag{4.1}$$

The JT action: a topological term with constant dilaton ϕ_0 (bulk piece $= \phi_0$ times the Euler characteristic of Eq. (4.3); boundary piece the Gibbons–Hawking–York term of Eq. (3.4)), plus a dynamical dilaton ϕ acting as a Lagrange multiplier that enforces constant negative curvature, $R = -2$ (locally AdS_2), on shell.

$$Z_{\mathrm{JT}} = e^{S_0\chi}\int \mathcal{D}g\,\mathcal{D}\phi\; e^{-S_{\mathrm{JT}}[g,\phi]} \tag{4.2}$$

The JT partition function (Euclidean continuation); the topological prefactor $e^{S_0\chi}$ organizes the sum over topologies by genus, with S_0 the extremal black hole entropy per unit Euler character.

$$\chi = 2 - 2g - b \tag{4.3}$$

The Euler characteristic of a genus-g surface with b boundary components; the disk ($g = 0$, $b = 1$) has $\chi = 1$.

$$\rho(E) = \frac{e^{S_0}}{2\pi^2}\sinh\!\Big(2\pi\sqrt{E}\Big) + \frac{1}{4\pi\sqrt{E}}\cos\!\Big(S_0 + \frac{\pi}{4} - 2\pi\sqrt{E}\Big) + \cdots \tag{4.4}$$

The leading terms of the non-perturbative JT spectral density. The first term is the disk-level Schwarzian density of states, $\rho_0(E) \propto \sinh(2\pi\sqrt{E})$; the second is a genus-dependent oscillatory correction of the type appearing in the exact random-matrix completion of JT gravity, where $\rho(E)$ is built order by order in the genus expansion of Eq. (15.4).

$$\langle Z(\beta)\rangle = \int_0^\infty dE\,\rho(E)\,e^{-\beta E} = \frac{e^{S_0}}{\sqrt{4\pi}\,\beta^{3/2}}\,e^{\pi^2/\beta} \tag{4.5}$$

The disk-level thermal partition function, obtained by Laplace-transforming the leading term of Eq. (4.4) and evaluating by saddle point; the enhancement $e^{\pi^2/\beta}$ at low temperature reflects the large density of near-extremal microstates.[2]

$$S_{\rm Sch} = -\phi_r \int d\tau\, \{f(\tau),\tau\}, \qquad \{f,\tau\} = \frac{f'''}{f'} - \frac{3}{2}\left(\frac{f''}{f'}\right)^2 \tag{4.6}$$

The Schwarzian action: the effective boundary theory obtained by integrating out the bulk of JT gravity (Eq. 4.1), governing a reparametrization $f(\tau)$ of the boundary time relative to the renormalized boundary coupling ϕ_r, with $\{f,\tau\}$ the Schwarzian derivative. This is precisely the low-energy effective action of the SYK model (Section 16) – the concrete statement underlying the JT/SYK duality of Eq. 16.4.

$$ds^2 = -\frac{r_+^2 - r^2}{\ell^2}\,\frac{du\,dv}{(1+uv)^2}\cdots, \qquad UV = -e^{2r_+ x/\ell^2} \tag{4.7}$$

The maximally extended (Kruskal-type) form of the eternal AdS–Schwarzschild geometry: analytically continuing past the horizon reveals a second, causally disconnected asymptotic region – the bulk dual of the second copy in the thermofield double state of Eq. 18.2, with the two exterior regions meeting at a spacelike wormhole (Einstein–Rosen bridge) that grows linearly in time, the geometric quantity conjectured to be dual to complexity in Eq. 20.3.

5. Chern–Simons Theory and Wilson Loop Observables

Three-dimensional Chern–Simons theory is exactly solvable and, via the Ashtekar–Barbero connection of Section 6, supplies the archetype for connection-based (as opposed to metric-based) quantization; it is also the historical seed of the loop-quantization program.

$$S_{\rm CS} = \frac{k}{4\pi}\int_M \mathrm{Tr}\left(A\wedge dA + \frac{2}{3}\,A\wedge A\wedge A\right) \tag{5.1}$$

The Chern–Simons action for a connection A valued in the Lie algebra of a gauge group G, at integer level k, on a 3-manifold M.

$$Z_{\rm CS}(S^3) = \sqrt{\frac{2}{k}}\,\sin\!\left(\frac{\pi}{k}\right) \tag{5.2}$$

[2] As supplied, the denominator was bracketed as $\sqrt{4\pi\beta^{3/2}}$, which would imply the weaker large-β falloff $\beta^{-3/4}$. Cross-checking against the primary literature (Stanford–Witten-type conventions), the correct saddle-point result scales as $\beta^{-3/2}$, consistent with reading the denominator as $\sqrt{4\pi}\,\beta^{3/2}$, as reproduced here. The exact $O(1)$ numerical prefactor is itself convention-dependent, varying with the normalization chosen for E and for the boundary value of the dilaton – published forms include $e^{S_0}e^{2\pi^2/\beta}/(\sqrt{2\pi}\,\beta^{3/2})$ under the alternative density-of-states convention $\rho_0(E)\propto\sinh(2\pi\sqrt{2E})$.

Witten's exact result for the $SU(2)$ Chern–Simons partition function on S^3.[3]

$$W_R(K) = \mathrm{Tr}_R\ \mathcal{P} \exp\left(\oint_K A\right) \tag{5.3}$$

The Wilson loop operator: the trace, in representation R, of the path-ordered holonomy of A around a contour K. Expectation values of products of such loops compute knot and link invariants (the Jones polynomial, for $SU(2)$).

$$\langle W_R(K)\rangle = \frac{\sin\!\left(\frac{\pi}{k}(2j+1)\right)}{\sin\!\left(\frac{\pi}{k}\right)} \tag{5.4}$$

The expectation value of an unknotted Wilson loop colored by the spin-j representation of $SU(2)$, subject to the same level-shift caveat as Eq. (5.2).

$$k \in \mathbb{Z} \tag{5.5}$$

The Chern–Simons level k is quantized to integer values: under a large (topologically nontrivial) gauge transformation the action of Eq. 5.1 shifts by $2\pi k \times$ (integer), so single-valuedness of $e^{iS_{\mathrm{CS}}}$ in the path integral forces $k \in \mathbb{Z}$ – a genuinely topological quantization condition, with no analogue among the continuously-tunable couplings of Sections 1–3.

$$c_{\mathrm{WZW}} = \frac{k\,\dim G}{k + h^\vee} \tag{5.6}$$

The level-rank matching between Chern–Simons theory on a manifold with boundary and the Wess–Zumino–Witten (WZW) model living on that boundary (Witten 1989): quantizing $SU(N)_k$ Chern–Simons theory on a solid handlebody produces the conformal blocks of the level-k WZW model, whose central charge is given here in terms of the dimension of the gauge group $\dim G$ and the dual Coxeter number $h^\vee$ already encountered in the level-shift footnote to Eq. 5.2.

$$K = 2\pi \int_{x>0} x\, T_{00}(x)\, d^{d-1}x \tag{5.7}$$

The (Bisognano–Wichmann) modular Hamiltonian for a Rindler wedge: the operator generating the same Rindler-boost flow whose associated temperature is the Unruh temperature above, and whose holographic generalization to an arbitrary entangling surface underlies entanglement wedge reconstruction – the statement that bulk operators in a wedge can be reconstructed from boundary data on the corresponding boundary region alone.

[3] Witten's 1989 derivation gives this result at the *shifted* level, $Z(S^3) = \sqrt{2/(k+2)}\,\sin[\pi/(k+2)]$; the shift $k \to k+2 = k + h^\vee$ (dual Coxeter number $h^\vee = 2$ for $SU(2)$) is a one-loop quantum effect, confirmed directly against the original result (Witten, *Commun. Math. Phys.* **121**, 351 (1989)), and is universal to every Wilson-loop and partition-function formula of $SU(2)_k$ Chern–Simons theory, including Eq. (5.4) below. Equation (5.2) as given is the unshifted, semiclassical-level expression.

6. BF Theory and the Plebanski–Holst Formulation of General Relativity

Casting general relativity as a constrained topological (BF) theory is the classical starting point for both spin foam models (Section 9) and the connection variables (Section 7) used in canonical loop quantum gravity.

$$\mathcal{L}_{\mathrm{BF}} = \frac{k}{2\pi} \int \mathrm{Tr}(B \wedge F) \tag{6.1}$$

The BF-theory Lagrangian: a 2-form B paired with the curvature 2-form F of a connection A (Eq. (6.2)). Unlike Chern–Simons or Einstein–Hilbert theory, BF theory is topological: it has no local propagating degrees of freedom until a simplicity constraint is imposed.

$$F = dA + A \wedge A \tag{6.2}$$

The curvature (field-strength) 2-form of a non-abelian connection A.

$$S_{\mathrm{Plebanski}} = \int \left[B^{IJ} \wedge F_{IJ} - \frac{1}{2}\, \phi_{IJKL}\, B^{IJ} \wedge B^{KL} \right] \tag{6.3}$$

The Plebanski action: BF theory for the internal (Lorentz or Euclidean) gauge group, with a Lagrange-multiplier field ϕ_{IJKL} (antisymmetric in each index pair, trace-free) enforcing the simplicity constraint that reduces the topological $B \wedge F$ theory on shell to general relativity in first-order (tetrad) form.

$$\phi_{IJKL}\, B^{IJ} \wedge B^{KL} = \Lambda\, \epsilon_{IJKL}\, B^{IJ} \wedge B^{KL} \tag{6.4}$$

The equation of motion obtained by varying ϕ_{IJKL}: the simplicity constraint, here with a cosmological constant Λ, solved (in its "gravitational" sector) by $B^{IJ} = e^I \wedge e^J$ for tetrad 1-forms e^I, which reproduces the Palatini action.

$$\mathcal{L}_{\mathrm{Holst}} = \frac{1}{16\pi G} \left[(e \wedge e)^* \wedge R + \frac{1}{\gamma}\, e \wedge e \wedge R \right] \tag{6.5}$$

The Holst action: the Palatini (tetrad) Lagrangian plus a term, weighted by the inverse Barbero–Immirzi parameter $1/\gamma$, that is topological (and torsion-free on shell) classically but becomes physical upon quantization – it fixes the overall scale of the LQG area and volume spectra.

$$A^{IJ}_{\mu} = \omega^{IJ}_{\mu} + \frac{1}{\gamma}\, \epsilon^{IJ}{}_{KL}\, \omega^{KL}_{\mu} \tag{6.6}$$

The Ashtekar–Barbero connection, built from the spin connection ω_μ^{IJ} and its internal dual, parametrized by γ; this $SU(2)$ connection is the canonical configuration variable of loop quantum gravity (Section 7).

$$E^\mu_I = \frac{1}{2}\,\epsilon^{IJKL}\,\epsilon^{\mu\nu\rho\sigma}\,e^J_\nu e^K_\rho e^L_\sigma \tag{6.7}$$

The densitized triad, the momentum conjugate to A_μ^{IJ}, built antisymmetrically from three factors of the tetrad e_μ^I.

$$D_{[\mu}e^I_{\nu]} = 0 \implies \omega = \omega(e), \qquad R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G\,T_{\mu\nu} \tag{6.8}$$

The field equations from independently varying the Palatini (first-order, tetrad-plus-connection) action with respect to ω and e: the first is the torsion-free condition, which (absent fermionic torsion sources) algebraically fixes ω to the usual metric-compatible spin connection built from e; substituting back reduces the second equation to the familiar second-order Einstein equations of Eq. 2.3.

$$h_e[A] = \mathcal{P}\exp\int_e A \tag{6.9}$$

The holonomy of the connection A along an edge e: the fundamental configuration-space observable of loop quantum gravity (whence its name), well defined non-perturbatively even though the connection itself is nowhere required to be smooth; spin-network states are built as functions of finitely many such holonomies, and the Hamiltonian constraint operator of Eq. 7.3 is regularized directly in terms of them.

7. Canonical Loop Quantum Gravity

Loop quantum gravity quantizes the phase space (A^i_a, E^a_i) non-perturbatively and without a background metric, in terms of holonomies of A along graphs and fluxes of E through surfaces. The classical constraint algebra of general relativity survives, in regularized form, as a triple of quantum operator constraints.

$$\mathcal{H}_{\mathrm{LQG}} = \frac{N}{\sqrt{\det E}}\Big[\epsilon_{ijk}F^i_{ab}E^a_jE^b_k - 2(1+\gamma^2)K^i_aK^j_bE^a_iE^b_j\Big] \;+\; N^a\Big[F^i_{ab}E^b_i - (1+\gamma^2)K^i_a\mathcal{G}_i\Big] \;+\; \lambda^i\mathcal{G}_i \tag{7.1}$$

The full smeared constraint of classical LQG, combining the scalar (Hamiltonian) constraint (lapse N), the vector (diffeomorphism) constraint (shift N^a), and the Gauss constraint (multiplier λ^i, Eq. (7.2)); the term proportional to $(1+\gamma^2)$ restores full Lorentzian covariance beyond the purely Euclidean piece $\epsilon_{ijk}F^iE^aE^b$.

$$\mathcal{G}_i = D_a E^a_i = \partial_a E^a_i + \epsilon_{ij}{}^k A^j_a E^a_k \tag{7.2}$$

The Gauss constraint, generating internal $SU(2)$ (frame) rotations; its vanishing is what makes E^a_i well defined independent of the choice of internal triad.

$$\hat{H}|\Psi\rangle = \sum_v \hat{H}_v|\Psi\rangle, \qquad \hat{H}_v = \frac{16\pi G}{\hbar}\sum_v \frac{N(v)}{\sqrt{\hat{V}_v}}\,\hat{C}_v \tag{7.3}$$

Thiemann's regularization of the quantum Hamiltonian constraint as a sum of vertex-supported operators $\hat{H}_v$ acting on spin-network states $|\Psi\rangle$, each weighted by the inverse square root of the volume operator at that vertex (Eq. (7.5)).

$$\hat{C}_v = \epsilon^{ijk}\hat{F}^i_{ab}\hat{E}^a_j\hat{E}^b_k - 2(1+\gamma^2)\hat{K}^i_a\hat{K}^j_b\hat{E}^a_i\hat{E}^b_j \tag{7.4}$$

The operator content of the vertex constraint $\hat{H}_v$: curvature and extrinsic-curvature contributions combined in the same Euclidean-plus-Lorentzian pattern seen classically above.

$$\hat{V}_v = \left(\frac{\beta}{3}\right)^{3/2}\sqrt{\left|\frac{i}{8\cdot 3!}\,\epsilon_{ijk}\,\epsilon^{abc}\,\hat{E}^a_i\hat{E}^b_j\hat{E}^c_k\right|} \tag{7.5}$$

The Ashtekar–Lewandowski volume operator evaluated at a vertex v, built from three flux operators; its discrete spectrum is, alongside the discrete area spectrum, one of the two headline kinematical predictions of LQG.

$$\hat{A}_S\,|j_1,\ldots,j_n\rangle = 8\pi\gamma\,\ell_{\mathrm{Pl}}^2\sum_{i=1}^{n}\sqrt{j_i(j_i+1)}\;|j_1,\ldots,j_n\rangle \tag{7.6}$$

The area operator spectrum: acting on a spin-network state whose edges $i=1,\ldots,n$ pierce a surface S, each carrying $SU(2)$ spin j_i, the area operator is diagonal with eigenvalues built from the Casimir $\sqrt{j_i(j_i+1)}$ at each puncture – alongside the volume spectrum of Eq. 7.5, the second headline discreteness prediction of LQG, and the microscopic origin of the Barbero–Immirzi-parameter-dependent counting behind the Bekenstein–Hawking entropy.

$$\Delta = 4\sqrt{3}\,\pi\gamma\,\ell_{\mathrm{Pl}}^2 \tag{7.7}$$

The area gap: the smallest possible nonzero eigenvalue of the area operator of Eq. 7.5′, obtained by setting a single puncture to its minimal spin $j=\frac{1}{2}$; it is this fundamental discreteness scale, not a free parameter, that regulates the LQC bounce of the modified Friedmann equation above and eliminates the classical singularity.

$$v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k = 0, \qquad z = \frac{a\,\dot{\phi}}{H} \tag{7.8}$$

The Mukhanov–Sasaki equation for the gauge-invariant curvature perturbation $v_k = z\,\mathcal{R}_k$: the master equation of cosmological perturbation theory, governing the quantum-to-classical evolution of the primordial fluctuations that seed CMB anisotropies and large-scale structure; replacing the classical background $a(t)$ by the LQC bounce of the effective Friedmann equation above extends this equation through the bounce, the standard route by which loop quantum cosmology makes contact with observational cosmology.

8. Loop Quantum Cosmology: Effective Semiclassical Dynamics

Loop quantum cosmology applies the LQG quantization scheme to minisuperspace (homogeneous, typically isotropic) cosmological models, most often using a massless scalar ϕ as an internal "clock." Its best-known result is the replacement of the classical Big Bang singularity by a quantum bounce.

$$\Delta\hat{p}_\phi^2 = \langle\hat{p}_\phi^2\rangle - \langle\hat{p}_\phi\rangle^2, \qquad \Delta\hat{V}^2 = \langle\hat{V}^2\rangle - \langle\hat{V}\rangle^2 \tag{8.1}$$

Quantum variances of the scalar-field momentum p_ϕ and of the volume V, used to quantify how sharply peaked (semiclassical) a given LQC state is.

$$\frac{d\langle\hat{V}\rangle}{d\phi} = \frac{\langle\hat{V}\hat{p}_\phi\rangle - \langle\hat{V}\rangle\langle\hat{p}_\phi\rangle}{\Delta p_\phi^2}\,\langle\hat{p}_\phi\rangle \tag{8.2}$$

The effective equation of motion for $\langle\hat{V}\rangle$ as a function of the internal time ϕ, obtained from Ehrenfest-type relations on a sharply peaked state; in the exactly solvable LQC models this yields a bounded, bouncing effective Friedmann equation in place of the classical singular one.

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_c}\right) \tag{8.3}$$

The effective (semiclassical) LQC Friedmann equation, with ρ_c a critical density of order the Planck density, set by the Barbero–Immirzi parameter and the minimum LQG area gap. The quantum-geometric correction $-\rho/\rho_c$, absent in classical cosmology, causes the expansion rate to vanish and reverse sign as $\rho \to \rho_c$ – the loop quantum bounce replacing the classical Big Bang singularity, following from the effective dynamics of Eq. 8.2 on FRW minisuperspace.

9. Spin Foam Models

Spin foams are the covariant (path-integral) counterpart to canonical LQG: a state sum over histories of spin networks, i.e. over 2-complexes labeled by $SU(2)$ representations on faces and intertwiners on edges.

$$\mathcal{H}_{\text{spin foam}} = \sum_{j_f, i_e} \prod_f (2j_f + 1) \prod_v A_v(j_f, i_e) \tag{9.1}$$

The spin foam state sum: a sum over labelings of faces f by spins j_f (weighted by the $SU(2)$ representation dimension $2j_f+1$) and edges e by intertwiners i_e, weighted by a vertex amplitude A_v at every vertex v of the 2-complex.

$$A_v(j_f, i_e) = \left\{ \begin{array}{ccc} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{array} \right\}_{i_e} \tag{9.2}$$

The vertex amplitude of the topological (BF-theory) spin foam model, given by the Wigner $6j$-symbol – the three-dimensional (Ponzano–Regge) case; four-dimensional gravitational models (EPRL/FK) modify this amplitude by imposing the simplicity constraint of Section 6 on the allowed representations.

$$\begin{aligned} \left\{ \begin{array}{ccc} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{array} \right\} = \sum_{m_i} (-1)^{\sum_{k=1}^{6}(j_k - m_k)} & \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_5 & j_6 \\ m_1 & -m_5 & m_6 \end{pmatrix} \\ & \times \begin{pmatrix} j_4 & j_2 & j_6 \\ m_4 & m_2 & -m_6 \end{pmatrix} \begin{pmatrix} j_4 & j_5 & j_3 \\ -m_4 & m_5 & m_3 \end{pmatrix} \end{aligned} \tag{9.3}$$

Racah's formula for the Wigner $6j$-symbol as a sum, over magnetic quantum numbers, of a product of four $3j$-symbols (themselves repackaged Clebsch–Gordan coefficients). Sign and phase conventions for this identity differ across standard references (e.g. Edmonds versus Varshalovich); the convention in use should be fixed explicitly wherever this formula feeds into an actual amplitude calculation.

$$A_v \sim N_+ \, e^{iS_{\text{Regge}}/\hbar} + N_- \, e^{-iS_{\text{Regge}}/\hbar} \tag{9.4}$$

The large-spin (semiclassical) asymptotics of the four-dimensional gravitational (EPRL/FK) spin foam vertex amplitude (Barrett *et al.*): as $j_f \to \infty$ at fixed ratios, the vertex amplitude oscillates as the cosine of the Regge action of Eq. 13.1 for the corresponding discrete geometry – the key consistency check that the covariant spin foam sum reproduces classical simplicial general relativity in the appropriate limit.

$$\rho_{\text{spec}}(\sigma) \xrightarrow{\sigma \to 0} \sim 2, \qquad \rho_{\text{spec}}(\sigma) \xrightarrow{\sigma \to \infty} \sim 4 \tag{9.5}$$

The spectral dimension flow in causal dynamical triangulations: probing the effective dimensionality of the quantum spacetime with a diffusion process of fictitious time σ reveals a UV spectral dimension close to 2 at short diffusion times, smoothly interpolating to the expected classical value of 4 at long diffusion times – numerically observed dynamical dimensional reduction, echoed independently in the asymptotic-safety and causal-set approaches to Sections 11–12.

$$a_{\mathrm{UV}} \geq a_{\mathrm{IR}} \tag{9.6}$$

The a-theorem: a distinguished coefficient a in the trace anomaly of an even-dimensional CFT decreases monotonically along any renormalization group flow from a UV to an IR fixed point – an irreversibility theorem constraining which pairs of CFTs can be connected by a flow at all, directly relevant to the search (Section 11) for a UV-complete flow ending at the asymptotic-safety fixed point g_* of Eq. 11.3.

10. Group Field Theory

Group field theory upgrades spin foam amplitudes to Feynman diagrams of a field theory on (several copies of) a group manifold, $\varphi : G^4 \to \mathbb{C}$ for four-dimensional models with $SU(2)$-valued arguments; individual quanta of φ represent tetrahedra, and the interaction vertex encodes their gluing into 4-simplices.

$$\begin{aligned}\mathcal{H}_{\mathrm{GFT}} = \int [dg]^4 \Big[& \frac{1}{2}\,\varphi(g_1,g_2,g_3,g_4)\,\mathcal{K}(g_i)\,\varphi(g_1,g_2,g_3,g_4) \\ & + \frac{\lambda}{5!}\,\varphi(g_1,g_2,g_3,g_4)\varphi(g_4,g_5,g_6,g_7)\varphi(g_7,g_3,g_8,g_9)\varphi(g_9,g_6,g_2,g_{10})\varphi(g_{10},g_8,g_5,g_1)\Big]\end{aligned} \tag{10.1}$$

The GFT free-energy functional: a quadratic (kinetic) term with kernel $\mathcal{K}$ plus a 5-valent (5!-symmetrized) interaction. Each field argument corresponds to a triangular face of a tetrahedron, and the index-contraction pattern glues five tetrahedra, sharing all ten triangles pairwise, into the boundary of a single 4-simplex.

$$\langle \mathcal{O} \rangle = \frac{1}{Z}\int \mathcal{D}\varphi\; \mathcal{O}\, e^{-S[\varphi]}, \qquad Z = \int \mathcal{D}\varphi\; e^{-S[\varphi]} \tag{10.2}$$

The Euclidean path-integral definition of expectation values in the GFT, formally identical to any Euclidean QFT.

$$\begin{aligned}S[\varphi] = \frac{1}{2}\int [dg]^4\,\varphi(g_i)\,\mathcal{K}(g_i)\,\varphi(g_i) \;+\; \frac{\lambda}{5!}\int [dg]^{10}\,\varphi(g_1,g_2,g_3,g_4)\varphi(g_4,g_5,g_6,g_7)\varphi(g_7,g_3,g_8,g_9) \\ \times\,\varphi(g_9,g_6,g_2,g_{10})\varphi(g_{10},g_8,g_5,g_1)\end{aligned} \tag{10.3}$$

The explicit GFT action reproducing the combinatorics of Eq. (10.1) as a local (in group space) field-theory action, with ten independent group integrations in the interaction term matching the ten shared triangles.

$$\mathcal{K}(g_i) = \sum_{j_i} \mathcal{K}_{j_i} \prod_{i=1}^{4} d_{j_i}\, \chi^{j_i}(g_i) \tag{10.4}$$

The kinetic kernel expanded in the Peter–Weyl (representation) basis: d_{j_i} are representation dimensions and χ^{j_i} the corresponding characters – the bridge between the group-field-theory formulation and the spin-network/spin-foam formulation of Section 9.

$$\Gamma_{\text{1PI}} = \sum_{\mathcal{G}} \frac{1}{\text{sym}(\mathcal{G})}\, \mathcal{A}_{\mathcal{G}}, \qquad \mathcal{A}_{\mathcal{G}} = \int [dg] \prod_{e} \delta(g_e) \prod_{f} \delta\left(\prod_{e \in \partial f} g_e \right) \tag{10.5}$$

The one-particle-irreducible effective action as a sum over GFT Feynman diagrams $\mathcal{G}$ (dual to simplicial complexes), each weighted by a symmetry factor and an amplitude enforcing, via delta functions, flatness of the holonomy around every face – reproducing, order by order, the spin-foam sum of Section 9.

$$S_{\text{Boulatov}}[\varphi] = \frac{1}{2} \int [dg]^3\, \varphi(g_1, g_2, g_3)^2 + \frac{\lambda}{4!} \int [dg]^6\, \varphi(g_1, g_2, g_3)\varphi(g_3, g_4, g_5)\varphi(g_5, g_2, g_6)\varphi(g_6, g_4, g_1) \tag{10.6}$$

The Boulatov model: the three-dimensional group field theory whose Feynman expansion reproduces exactly the Ponzano–Regge (topological BF-theory) spin foam amplitudes of Eqs. 9.2–9.3 – historically the first GFT, and the template later generalized to the four-dimensional models discussed in Section 10.

11. Asymptotic Safety

The asymptotic safety program seeks a non-perturbative, non-Gaussian UV fixed point of the gravitational renormalization group flow, which would render gravity predictive as an ordinary (if not perturbatively renormalizable) Wilsonian quantum field theory, without new degrees of freedom or a discretization of spacetime.

$$\mathcal{L}_{\text{asymptotic safety}} = \sqrt{-g} \left[\frac{1}{16\pi G} R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \gamma R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} + \cdots \right] \tag{11.1}$$

A truncation of the gravitational effective average action Γ_k (scale-dependent through an infrared cutoff k), structurally identical to the higher-curvature Lagrangian of Section 3, but here every coupling $G, \alpha, \beta, \gamma, \ldots$ is understood to run with k according to the functional renormalization group equation.

$$\beta_g = \mu \frac{dg}{d\mu} = (2 - d)g + \frac{g^2}{16\pi^2} b_1 + \frac{g^3}{(16\pi^2)^2} b_2 + \cdots \tag{11.2}$$

The beta function for a generic coupling g in d dimensions: the classical (dimensional) scaling term $(2-d)g$ plus perturbative loop corrections $b_1, b_2, \ldots$; this schematic perturbative form is what non-perturbative functional-RG methods extend into the strong-coupling regime where the fixed point of interest is expected to live.

$$g_* : \ \beta_g(g_*) = 0 \tag{11.3}$$

The defining condition for an RG fixed point. The asymptotic-safety conjecture is that gravity possesses a non-Gaussian fixed point ($g_* \neq 0$) with a finite-dimensional UV-critical surface, so that the continuum limit $k \to \infty$ can be taken with only finitely many free parameters.

$$k\,\partial_k \Gamma_k = \frac{1}{2}\,\mathrm{Tr}\left[\left(\Gamma_k^{(2)} + R_k\right)^{-1} k\,\partial_k R_k\right] \tag{11.4}$$

The Wetterich equation: the exact functional renormalization group flow equation for the effective average action Γ_k, with $\Gamma_k^{(2)}$ its second functional derivative and R_k an infrared regulator suppressing modes below scale k. This is the master equation of the asymptotic safety program: truncating Γ_k to the form of Eq. 11.1 and inserting it here generates the beta functions of Eq. 11.2 non-perturbatively.

12. Causal Set Theory

Causal set theory takes seriously the idea that spacetime is fundamentally a locally finite partially ordered set: discreteness supplies an intrinsic UV cutoff, and the partial order alone is postulated to encode, in the continuum limit, the full causal structure of a Lorentzian manifold.

$$\mathcal{L}_{\text{causal sets}} = \sum_{x,y \in \mathcal{C}} \mathcal{K}(x,y)\,\phi(x)\phi(y) + \lambda \sum_{x \in \mathcal{C}} \phi(x)^4 \tag{12.1}$$

A discretized scalar field action on a causal set $\mathcal{C}$: the kinetic term is built from a causal-set analogue of the continuum kinetic operator (a discrete, non-local kernel $\mathcal{K}(x,y)$ constructed from the order relation alone), plus a local quartic self-interaction.

$$\langle \phi(x)\phi(y) \rangle = \frac{1}{\mathcal{K}(x,y)} \tag{12.2}$$

The free (Gaussian) two-point function as the matrix inverse of the kinetic kernel – the causal-set discretization of the general fact that, for any quadratic action, the propagator is the inverse of the kinetic operator (cf. Eq. 1.3 in the continuum limit).

$$S_{\mathrm{BD}} = \zeta_d \left(N + \frac{\beta_d}{\alpha_d} \sum_i C_i^{(d)}\, N_i \right) \tag{12.3}$$

The Benincasa–Dowker–Glaser discrete action: N is the cardinality of the causal set and N_i the number of i-inclusive order intervals (pairs of elements with exactly $i-1$ causet elements causally between them); $\zeta_d, \alpha_d, \beta_d$, and the rational coefficients $C_i^{(d)}$ are dimension-dependent constants fixed by requiring that the continuum limit, on a causal set obtained by Poisson-sprinkling into a curved Lorentzian manifold, converges to the Einstein–Hilbert action – the causal-set analogue of the discretized Regge action of Eq. 13.1.

$$P(n \text{ points in } V) = \frac{(\rho V)^n}{n!}\, e^{-\rho V} \tag{12.4}$$

Poisson sprinkling: the procedure by which a causal set is generated from a continuum Lorentzian manifold, scattering points via a Poisson process of density ρ, so the expected number of points in any region equals ρ times its invariant spacetime volume V – the unique sprinkling measure that is both Lorentz invariant and reproduces the correct element density in every reference frame, unlike a naive regular lattice.

13. Simplicial Quantum Gravity: Regge Calculus and Dynamical Triangulations

Simplicial quantum gravity discretizes the path integral itself, replacing smooth metrics by piecewise-flat triangulations built from flat simplices glued along their faces, with curvature concentrated – as a deficit angle – on the codimension-2 hinges.

$$S_{\text{Bergmann}} = \sum_{4\text{-simplices}} \left(\sum_{\text{edges}} l_e^2\, \varepsilon_e + \lambda V_4 \right) \tag{13.1}$$

A simplicial gravitational action of Regge type: squared edge lengths l_e^2 weighted by deficit angles ε_e, plus a cosmological-constant term proportional to the total 4-volume V_4.[4]

$$Z = \int \mathcal{D}l_e\; e^{iS[l_e]} \tag{13.2}$$

The (Lorentzian) quantum Regge calculus path integral: an integral over the edge lengths l_e, the fundamental variables of the simplicial geometry, in place of $\mathcal{D}g_{\mu\nu}$.

$$\mathcal{L}_{\text{dynamical triangulations}} = \sum_T \frac{1}{C_T}\, e^{-S_E[T]}, \qquad S_E[T] = \kappa_4 N_4 - \kappa_2 N_2 \tag{13.3}$$

[4]The standard name for this construction is the *Regge action* (T. Regge, 1961), $S_{\text{Regge}} = \sum_{\text{hinges}} A_h\, \varepsilon_h - \Lambda \sum_\sigma V_\sigma$: the deficit angle multiplies the volume A_h of the codimension-2 *hinge* on which curvature is concentrated. In four dimensions the hinges are triangles and A_h is a triangle area; in three dimensions the hinges are the edges themselves and $A_h = l_e$, linear in the edge length. Neither case produces the squared edge length l_e^2 appearing above; it is reproduced here exactly as supplied.

The (causal or Euclidean) dynamical-triangulations discretized path-integral weight for a triangulation T: C_T is the order of its automorphism group, and the discretized Einstein–Hilbert action $S_E[T]$ is written directly in terms of the number of 4-simplices N_4 (bare cosmological term) and the number of triangles N_2 (bare inverse Newton constant) – all edge lengths are fixed to a single value, so curvature information is carried by the triangulation's combinatorics alone.

$$Z = \sum_{T \in \mathcal{T}} \frac{1}{C_T}\, e^{-S_E[T]} \tag{13.4}$$

The dynamical-triangulations partition function: a sum over the ensemble $\mathcal{T}$ of triangulations of fixed topology.

$$\varepsilon_h = 2\pi - \sum_{\sigma \supset h} \theta_\sigma(h) \tag{13.5}$$

The deficit angle at a hinge h: the shortfall of the sum of dihedral angles $\theta_\sigma(h)$ contributed by every 4-simplex σ meeting at h, from the flat-space value 2π. A nonzero ε_h is the simplicial-geometry manifestation of curvature concentrated at h, and is exactly the quantity entering the Regge action of Eq. 13.1.

14. Random Tensor Models

Random tensor models generalize the matrix models of Section 15 to rank $d > 2$: the Feynman expansion of a tensor field theory naturally generates – and, in a suitable $1/N$ expansion, is dominated by – triangulations of d-dimensional pseudomanifolds, providing a higher-dimensional analogue of the two-dimensional topological expansion.

$$\mathcal{L}_{\text{tensor models}} = \sum_{a_1,\ldots,a_d} T_{a_1\ldots a_d}\bar{T}^{a_1\ldots a_d} - \frac{\lambda}{N^{d(d-1)/4}} \sum_{a,b} T_{a_1\ldots a_d}\bar{T}^{b_1\ldots b_d}\, \delta_{a_1 b_1}\cdots\delta_{a_d b_d} \tag{14.1}$$

The action of a complex rank-d tensor model: a Gaussian kinetic term plus a single interaction of "melonic" type, with N-scaling tuned so that a non-trivial large-N limit exists, dominated by the maximally singular (melonic) Feynman graphs.

$$Z = \int dT\, d\bar{T}\; e^{-S[T,\bar{T}]} \tag{14.2}$$

The tensor-model partition function; its melonic large-N limit is the higher-dimensional generalization of the matrix-model genus expansion of Eq. (15.4), and is related, through the same index-contraction combinatorics, to the SYK model of Section 16.

$$F = \sum_{\omega=0}^{\infty} N^{\,d-1-c_d\,\omega}\, F_\omega, \qquad c_d > 0 \tag{14.3}$$

The $1/N$ expansion of the tensor-model free energy, organized by the non-negative integer degree ω (a combinatorial invariant of each Feynman graph, generalizing the matrix-model genus of Eq. 15.4). Gurau's theorem guarantees $c_d > 0$, so the $\omega = 0$ ("melonic") graphs dominate as $N \to \infty$ – exactly analogous to how planar ($g = 0$) graphs dominate the matrix-model expansion below.

15. Random Matrix Models and Two-Dimensional Gravity

Hermitian one-matrix models, in their 't Hooft ($N \to \infty$) limit, sum precisely over triangulated two-dimensional surfaces weighted by the discretized Einstein–Hilbert action – the original example of an exactly solvable discretized quantum gravity, and, as noted in Section 4, the exact non-perturbative completion of JT gravity.

$$\mathcal{L}_{\text{matrix models}} = \operatorname{Tr}\left(\frac{1}{2}M^2 + \frac{g}{3}M^3\right) \tag{15.1}$$

A cubic single-trace Hermitian matrix model; the cubic (or, for stability, even) interaction generates a genus expansion whose leading diagrams triangulate the sphere.

$$Z = \int dM\; e^{-N \operatorname{Tr} V(M)} \tag{15.2}$$

The matrix-model partition function, with 't Hooft scaling N in the exponent so that the genus expansion of Eq. (15.4) organizes as a power series in $1/N^2$.

$$\rho(\lambda) = \frac{1}{2\pi}\sqrt{4 - \lambda^2} \tag{15.3}$$

The Wigner semicircle law: the large-N eigenvalue density of a Gaussian Hermitian random matrix, support normalized to $[-2, 2]$ – the matrix-model analogue of the free-fermion/SYK spectral densities of Sections 4 and 16.

$$F = \sum_{g=0}^{\infty} N^{2-2g} F_g \tag{15.4}$$

The 't Hooft topological (genus) expansion of the matrix-model free energy: F_g is the contribution of genus-g surfaces, weighted by N^{χ} with $\chi = 2 - 2g$ the Euler characteristic of Eq. (4.3) at $b = 0$.

$$W(x) = \frac{1}{N}\left\langle \operatorname{Tr}\frac{1}{x - M}\right\rangle \xrightarrow{N\to\infty} \int d\lambda\, \frac{\rho(\lambda)}{x - \lambda} \tag{15.5}$$

The resolvent (Stieltjes transform) of the matrix model: at large N it is determined by a set of loop (Schwinger–Dyson) equations following from invariance of Z under $M \to M + \delta M$, and its discontinuity across the eigenvalue support recovers the spectral density $\rho(\lambda)$ of Eq. 15.3 – the standard route by which matrix models are solved exactly at large N.

$$\lambda = g^2 N \tag{15.6}$$

The 't Hooft coupling: holding λ fixed while $N \to \infty$ (the 't Hooft limit) is what makes the genus expansion of Eq. 15.4 a controlled double expansion – in $1/N$ at fixed λ, and then in λ – the same limit underlying planar-diagram dominance throughout large-N gauge theory, and, via AdS/CFT (Section 17), the weak-coupling regime of the dual bulk string theory.

$$\mathcal{F}(t) \equiv \frac{1}{N} \sum_i \left\langle [\psi_i(t), \psi_i(0)]^2 \right\rangle \sim -\frac{1}{N}\, e^{\lambda_L t} \tag{15.7}$$

The out-of-time-order correlator (OTOC) that operationally defines the Lyapunov exponent λ_L of the MSS bound above: its exponential growth (up to the scrambling time $t_* \sim \frac{\beta}{2\pi} \log N$, after which the $1/N$ perturbative expansion itself breaks down) is the sharp, calculable diagnostic of quantum chaos that both SYK and black holes exhibit at maximal strength.

$$c_L = 12\, J \tag{15.8}$$

The Kerr/CFT central charge: the near-horizon geometry of an extremal Kerr black hole of angular momentum J has an enhanced conformal symmetry, and applying the Cardy formula above with this central charge reproduces the Bekenstein–Hawking entropy of the extremal Kerr black hole – a rotating-black-hole echo of the BTZ/Cardy matching in Section 17, extending microscopic entropy counting beyond the AdS_3 examples where it was first established.

16. The Sachdev–Ye–Kitaev Model and Its Gravitational Dual

The SYK model – N Majorana fermions with random, all-to-all q-body interactions – is solvable at large N, develops an emergent conformal symmetry at strong coupling, and saturates the conjectured bound on Lyapunov chaos, which together make it the leading microscopic model dual to JT gravity (Section 4) in the near-AdS_2 throat of a near-extremal black hole.

$$\mathcal{L}_{\text{SYK}} = \sum_i \psi_i\, \partial_\tau \psi_i + \sum_{i<j<k<l} J_{ijkl}\, \psi_i \psi_j \psi_k \psi_l \tag{16.1}$$

The SYK Lagrangian for N Majorana fermions ψ_i ($q = 4$ case), with independent random Gaussian couplings J_{ijkl} of zero mean and variance $\propto J^2/N^3$.

$$\langle \psi_i(\tau) \psi_j(0) \rangle = \frac{b}{|\tau|^{2\Delta}}\, \mathrm{sgn}(\tau) \tag{16.2}$$

The emergent conformal (IR) two-point function, with scaling dimension $\Delta = 1/q = 1/4$ for the $q = 4$ model – the hallmark of the strongly coupled, reparametrization-invariant regime whose residual (Schwarzian) dynamics matches the boundary dynamics of JT gravity.

$$G(\tau)^{-1} = -\partial_\tau - \Sigma(\tau), \qquad \Sigma(\tau) = J^2\, G(\tau)^3 \tag{16.3}$$

The large-N, disorder-averaged Schwinger–Dyson equations for the SYK Green's function G and self-energy Σ; the power $G^3 = G^{q-1}$ reflects the four-fermion ($q = 4$) interaction.

$$\mathcal{L}_{\mathrm{JT/SYK}} = \frac{\phi_0}{16\pi G}\left(\int d^2x\sqrt{-g}R + 2\int_\partial K\right) + \frac{1}{16\pi G}\int d^2x\sqrt{-g}\,\phi\,(R+2) + S_{\mathrm{SYK}}[\chi_i] \tag{16.4}$$

The combined bulk-plus-boundary description used to exhibit the JT/SYK duality: the JT action of Eq. (4.1) supplemented by an independent copy of the SYK action for boundary degrees of freedom χ_i.

$$Z_{\mathrm{JT/SYK}} = \int \mathcal{D}g\,\mathcal{D}\phi\,\mathcal{D}\chi_i\; e^{-S_{\mathrm{JT}}[g,\phi]-S_{\mathrm{SYK}}[\chi_i]} \tag{16.5}$$

The corresponding combined partition function; quantitatively matching the two sides (e.g. comparing Eq. (4.4) to the SYK many-body density of states) is among the sharpest quantitative tests of holography available outside of string theory.

$$\lambda_L \le \frac{2\pi}{\beta} \tag{16.6}$$

The Maldacena–Shenker–Stanford (MSS) bound on quantum chaos: the Lyapunov exponent λ_L governing the exponential growth of an out-of-time-order correlator cannot exceed $2\pi/\beta$ for any thermal quantum system. The SYK model saturates this bound at strong coupling, exactly as a black hole horizon does – the sharpest evidence that SYK is "maximally chaotic" in the black-hole sense.

$$\partial_X S_{\mathrm{gen}}(X) = 0, \qquad S_{\mathrm{gen}} = \frac{\mathrm{Area}(X)}{4G} + S_{\mathrm{bulk}} \tag{16.7}$$

The quantum extremal surface (QES) condition: extremizing the generalized entropy S_{gen} (area plus bulk entanglement entropy) over the choice of bulk surface X homologous to a boundary region – the precise extremization that produces both the RT surface γ_A of Eq. 19.2 at leading order and the island contributions of Eq. 19.4 once bulk quantum corrections dominate.

$$\sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}}\,\mathcal{G}_{\Delta,\ell}(u,v) = \left(\frac{u}{v}\right)^{\Delta_\phi}\sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}}\,\mathcal{G}_{\Delta,\ell}(v,u) \tag{16.8}$$

The conformal bootstrap crossing equation: demanding that the four-point function $\langle\phi\phi\phi\phi\rangle$ of a CFT primary ϕ give the same answer whether expanded in the (12)(34) or (14)(23) operator product channel constrains the spectrum of operator dimensions Δ, spins ℓ, and OPE coefficients $\lambda_{\phi\phi\mathcal{O}}$ purely from consistency, with no reference to any Lagrangian – the modern, non-perturbative approach to solving strongly coupled CFTs (and, via AdS/CFT, strongly coupled quantum gravity) that has produced the most precise known determination of the 3d Ising model's critical exponents.

$$S_{\mathrm{rel}}(\rho_A\|\sigma_A) = \Delta\langle H_{\mathrm{bulk}}\rangle \tag{16.9}$$

The JLMS formula (Jafferis–Lewkowycz–Maldacena–Suh): the boundary relative entropy between two nearby states, restricted to region A, equals the relative entropy of the corresponding bulk states in A's entanglement wedge – the precise, quantitative statement that makes bulk reconstruction in AdS/CFT a special case of a quantum error-correcting code, with the entanglement wedge playing the role of the code subspace protected against erasure of the complementary boundary region.

17. Holography: The dS/CFT and AdS/CFT Correspondences

Holographic dualities equate a $(d{+}1)$-dimensional gravitational bulk theory with a d-dimensional conformal field theory living on its boundary. AdS/CFT is by far the best established instance, with an explicit string-theoretic derivation in specific cases; dS/CFT extends the same dictionary, more speculatively, to de Sitter asymptotics.

$$\mathcal{L}_{\mathrm{dS/CFT}} = \int_{\partial\,\mathrm{dS}} d^3x\,\sqrt{\gamma}\left[\frac{1}{2}\phi(-\Box + m^2)\phi + \frac{\lambda}{3!}\phi^3\right] \tag{17.1}$$

A putative boundary CFT action dual to a bulk scalar under dS/CFT, structurally parallel to the AdS/CFT example of Eq. (17.3) but evaluated on the (spacelike) conformal boundary of de Sitter space rather than the timelike boundary of AdS.

$$\langle\mathcal{O}(x_1)\cdots\mathcal{O}(x_n)\rangle_{\mathrm{CFT}} = \left.\frac{\delta^n Z_{\mathrm{dS}}}{\delta\phi_0(x_1)\cdots\delta\phi_0(x_n)}\right|_{\phi_0=0} \tag{17.2}$$

The dS/CFT correlator-generating prescription: boundary correlators of the dual operator $\mathcal{O}$ are functional derivatives of the bulk (dS) partition function with respect to the boundary value ϕ_0 of the bulk field – the de Sitter analogue of the Gubser–Klebanov–Polyakov/Witten relation of Eq. (17.4).

$$\mathcal{L}_{\mathrm{AdS/CFT}} = \int_{\mathrm{AdS}} d^{d+1}x\,\sqrt{g}\left[\frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{3!}\phi^3\right] \tag{17.3}$$

The canonical textbook example of a bulk AdS scalar field action with cubic self-interaction, whose boundary correlators (via Eq. (17.4)) reproduce a boundary CFT with an operator $\mathcal{O}$ of dimension Δ fixed by $m^2\ell^2 = \Delta(\Delta - d)$.

$$\left\langle e^{\int d^d x\, \phi_0 \mathcal{O}} \right\rangle_{\text{CFT}} = Z_{\text{bulk}}[\phi \to \phi_0] \tag{17.4}$$

The Gubser–Klebanov–Polyakov/Witten master relation: the CFT generating functional of connected correlators of $\mathcal{O}$, sourced by the boundary value ϕ_0, equals the bulk partition function evaluated with Dirichlet boundary condition $\phi \to \phi_0$.

$$c = \frac{3\ell}{2G} \tag{17.5}$$

The Brown–Henneaux central charge: for three-dimensional AdS gravity of radius ℓ, the asymptotic symmetry algebra of AdS_3 is generated by two copies of the Virasoro algebra with this central charge – historically the first (1986, pre-Maldacena) hint of a holographic CFT dual to a gravitational theory, recovered as a special case of the AdS/CFT dictionary of Eq. 17.4.

$$S = 2\pi\sqrt{\frac{c\,L_0}{6}} \tag{17.6}$$

The Cardy formula: the asymptotic density of states of a unitary two-dimensional CFT at large energy (Virasoro zero-mode eigenvalue L_0), fixed entirely by the central charge c. Applied with the central charge $c = 3\ell/2G$ of the previous entry (Brown–Henneaux), the Cardy formula reproduces the Bekenstein–Hawking entropy of the BTZ black hole exactly – one of the first fully microscopic derivations of black hole entropy from a dual quantum theory, predating the D-brane counting of higher-dimensional black holes.

$$\Delta\mathcal{C} \sim -k\,\big(\text{number of shocks}\big) \tag{17.7}$$

The switchback effect: a precisely timed perturbation (shock wave) inserted into an eternal black hole's interior *delays* the linear growth of complexity of Eq. 20.3 by an amount independent of how long ago the perturbation was applied – a distinctively gravitational, non-local-in-time signature of complexity growth with no obvious analogue in a generic chaotic quantum system, and a nontrivial check of the complexity conjectures.

18. Thermofield Double States and Eternal Black Holes

The thermofield double construction gives a pure-state, unitary description of a thermal ensemble by entangling two copies of a system; holographically it is dual to the two-sided eternal AdS black hole, and, once weakly coupled, to a traversable wormhole.

$$\mathcal{L}_{\text{TFD}} = \mathcal{L}_L + \mathcal{L}_R + \lambda \sum_i \mathcal{O}_L^i \mathcal{O}_R^i \tag{18.1}$$

Two decoupled copies of a CFT (left, L, and right, R) plus a direct coupling between operators on the two sides. A coupling of exactly this form (small, relevant, and of the correct sign) is what renders the eternal black hole's wormhole traversable (Gao–Jafferis–Wall); it is an addition made *on top of* the thermofield double construction, not part of its definition.

$$|\mathrm{TFD}\rangle = \frac{1}{\sqrt{Z}} \sum_n e^{-\beta E_n/2} |n\rangle_L |n\rangle_R \tag{18.2}$$

The thermofield double state: purifying the thermal density matrix $\rho = e^{-\beta H}/Z$ of a single copy by entangling it with an identical auxiliary copy; tracing out either side reproduces the thermal ensemble on the other.

$$T_{\mathrm{Unruh}} = \frac{\hbar a}{2\pi} \tag{18.3}$$

The Unruh temperature: an observer with uniform proper acceleration a in the Minkowski vacuum perceives a thermal bath at this temperature; restricting the vacuum to a single Rindler wedge produces an entangled state of exactly the thermofield-double form of Eq. 18.2, with the two Rindler wedges playing the role of the L, R copies – the flat-space precursor of the eternal-black-hole/TFD correspondence.

$$S_L = -\operatorname{Tr}(\rho_L \ln \rho_L) = -\sum_n p_n \ln p_n, \qquad p_n = \frac{e^{-\beta E_n}}{Z} \tag{18.4}$$

The entanglement entropy of one side of the thermofield double, obtained by tracing Eq. 18.2 over the R copy: it coincides exactly with the ordinary thermal (Gibbs) entropy of a single copy at temperature $1/\beta$ – the simplest illustration of the principle, central to this section, that horizon/thermal entropy is a genuine entanglement entropy once a purifying reference system is available.

$$\mathcal{N}(\rho) = \frac{\left\|\rho^{T_B}\right\|_1 - 1}{2} \tag{18.5}$$

The entanglement negativity: built from the trace norm of the partial transpose ρ^{T_B}, this is a genuine entanglement measure (unlike the mutual information $I(A:B)$ of Eq. 19.3) even for *mixed* states – relevant precisely when A and B are two disjoint boundary regions embedded in a larger, entangled system, and holographically dual to the area of a different bulk surface (the entanglement wedge cross-section) than the RT surface γ_A of Eq. 19.2 alone can capture.

19. Holographic Entanglement Entropy, Islands, and the Page Curve

The Ryu–Takayanagi proposal and its quantum-corrected (island) generalization identify entanglement entropy in the boundary theory with the area of an extremal bulk surface, providing

the sharpest known dictionary between spacetime geometry and quantum information – and the tool that finally reconciled black hole evaporation with unitarity.

$$S_{\mathrm{EE}} = \frac{\mathrm{Area}(\gamma_A)}{4G} + S_{\mathrm{bulk}} \tag{19.1}$$

The quantum-corrected (Faulkner–Lewkowycz–Maldacena) holographic entanglement entropy of a boundary region A: the area of an extremal bulk surface γ_A homologous to A, plus the bulk entanglement entropy S_{bulk} of quantum fields between γ_A and A.

$$\gamma_A = \partial(\text{entanglement wedge}) \tag{19.2}$$

The RT/HRT surface, identified as (the bulk portion of) the boundary of the entanglement wedge of A – the bulk domain of dependence reconstructible from boundary data on A alone.

$$I(A:B) = S(A) + S(B) - S(A \cup B) \tag{19.3}$$

The mutual information between two boundary regions A, B; manifestly UV-finite (divergences in $S(A)$ and $S(B)$ cancel against those in $S(A \cup B)$), and an upper bound on correlators between operators in A and B.

$$S_{\mathrm{island}} = \min \operatorname{ext}\left[\frac{\mathrm{Area}(\partial I)}{4G} + S_{\mathrm{bulk}}(R \cup I)\right] \tag{19.4}$$

The island formula (Almheiri–Engelhardt–Marolf–Wall; Penington): the generalized entropy of a radiation region R, extremized over the choice of an "island" region I that may lie behind the horizon, minimized over extrema. At early times the trivial island ($I = \varnothing$) dominates, reproducing Hawking's original ever-increasing result; after the Page time a non-trivial island dominates, capping the entropy and producing the unitary Page curve of Eq. (19.5).

$$S_{\mathrm{Page}}(t) = \begin{cases} S_{\mathrm{radiation}}(t), & t < t_{\mathrm{Page}} \\ S_{\mathrm{black\ hole}}(t), & t > t_{\mathrm{Page}} \end{cases} \tag{19.5}$$

The Page curve: entanglement entropy of the radiation tracks the growing coarse-grained radiation entropy before the Page time, and the shrinking black hole entropy after – the qualitative shape that Eq. (19.4) reproduces from a genuine bulk calculation.

$$t_{\mathrm{Page}} \sim \frac{S_{\mathrm{BH}}}{T} \tag{19.6}$$

An order-of-magnitude estimate of the Page time: comparable to the black hole's Hawking evaporation timescale, i.e. a finite fraction of the total evaporation time rather than a parametrically shorter or longer scale.

$$S(\rho) = -\lim_{n\to 1} \partial_n \log \operatorname{Tr} \rho^n \tag{19.7}$$

The replica trick: the von Neumann entropy is obtained by analytically continuing the more tractable Rényi entropies $S_n = \frac{1}{1-n}\log \operatorname{Tr}\rho^n$ to $n \to 1$. Gravitationally, computing $\operatorname{Tr}\rho^n$ via an n-fold replicated path integral and extremizing over bulk saddles is precisely what produces the island contributions to Eq. 19.4 ("replica wormholes"), the modern derivation of the Page curve of Eq. 19.5.

$$P_{\mathrm{GW}} = \frac{G}{5c^5}\left\langle \dddot{Q}_{ij}\, \dddot{Q}^{ij} \right\rangle \tag{19.8}$$

The (Einstein) quadrupole formula for gravitational-wave luminosity, in terms of the traceless mass quadrupole moment Q_{ij}: the leading weak-field, slow-motion prediction of linearized Eq. 2.3, and the formula whose direct experimental confirmation – via the orbital decay of the Hull–Taylor binary pulsar, and later the LIGO/Virgo detections themselves – is the most stringent quantitative test to date of general relativity in the regime this entire Part is built to extend.

20. Holographic Complexity

Quantum computational complexity – roughly, the minimal number of elementary gates needed to prepare a target state from a reference state – has been proposed as the quantity holographically dual to the growth of the wormhole interior long after the entanglement-entropy probe of Eq. (19.1) (Section 19) has saturated at its Page-time plateau.

$$\mathcal{L}_{\text{complexity}} = \int d\tau \ \operatorname{Tr}\!\left(U^{\dagger}\frac{dU}{d\tau} H \right) \tag{20.1}$$

A control-theoretic cost functional for a unitary circuit $U(\tau)$ generated by a (possibly τ-dependent) Hamiltonian $H(\tau) = i\,\frac{dU}{d\tau}U^{-1}$; in Nielsen's geometric approach to complexity, H plays the role of a "velocity" on the space of unitaries.

$$\mathcal{C} = \min_{U} \int_0^1 d\tau \ F(U, \dot{U}) \tag{20.2}$$

Nielsen complexity: the length, in an appropriately chosen right-invariant (Finsler or Riemannian) metric F on the unitary group, of the shortest path from the identity to the target unitary – recasting circuit complexity as a geodesic-distance problem.

$$\mathcal{C}_{\text{volume}} = \frac{V}{G\ell}, \qquad \mathcal{C}_{\text{action}} = \frac{S}{\pi\hbar} \tag{20.3}$$

The two leading holographic complexity conjectures: "complexity=volume" (Susskind), identifying $\mathcal{C}$ with the volume V of a maximal-volume bulk Cauchy slice anchored on the boundary time slice, divided by $G\ell$; and "complexity=action" (CA), identifying $\mathcal{C}$ with the on-shell gravitational action S evaluated on the Wheeler–DeWitt patch, divided by $\pi\hbar$. Both are conjectured to grow linearly for a time far exceeding the Page/scrambling time of Section 19, matching the expected late-time linear growth of circuit complexity.

$$\frac{d\mathcal{C}}{dt} \leq \frac{2E}{\pi\hbar} \tag{20.4}$$

The Lloyd bound: the rate of computation (complexity growth) of any physical system is bounded by its energy E above the ground state. Holographic complexity growth (Eq. 20.3) is conjectured to saturate this bound at late times – a further quantitative check that the complexity=action proposal describes a genuine computational process rather than an ad hoc geometric quantity.

Part II

Part II — Quantum and Computational Chemistry

Preface to Part II: Scope and Conventions

This second part turns from quantum gravity to quantum and computational chemistry: the hierarchy of approximations by which the many-electron Schrödinger equation is made tractable enough to predict molecular structure, spectra, reactivity, and thermodynamic properties. Thirteen sections proceed roughly in the order of a graduate electronic-structure course: the Born–Oppenheimer separation (Section 21); wavefunction-based methods of increasing sophistication – Hartree–Fock (Section 22), basis sets (Section 23), Møller–Plesset perturbation theory (Section 24), configuration interaction and coupled cluster (Section 25); density-functional methods for ground states (Section 26) and excited states (Sections 27–28); the spectroscopy and statistical mechanics built on top of an electronic structure calculation (Sections 29–30); chemical kinetics and electrochemistry (Sections 31–32); and, finally, the second-quantized and qubit reformulations of the electronic-structure problem underlying quantum computing for chemistry (Section 33), which closes the compendium by returning to the quantum-information language of Part I.

Conventions. Atomic units ($\hbar = m_e = e = 4\pi\varepsilon_0 = 1$) are used throughout, as is standard in the electronic-structure literature; energies are then in hartree (E_h). Molecular spin-orbitals are denoted $\chi_p(\mathbf{x})$ (space-spin coordinate $\mathbf{x}$), with indices i, j, k, l reserved for occupied and a, b, c, d for virtual (unoccupied) orbitals, and p, q, r, s for either; a spin-orbital formulation is used unless a spatial-orbital (restricted, closed-shell) formulation is stated explicitly. Two-electron integrals are written in physicist's notation, $\langle pq|rs\rangle = \int d\mathbf{x}_1 d\mathbf{x}_2\, \chi_p^*(\mathbf{x}_1)\chi_q^*(\mathbf{x}_2)\, r_{12}^{-1}\, \chi_r(\mathbf{x}_1)\chi_s(\mathbf{x}_2)$, with antisymmetrized combination $\langle pq\|rs\rangle = \langle pq|rs\rangle - \langle pq|sr\rangle$; this (rather than the equally common "chemist's" $(pq|rs)$ notation) is the convention of Szabo & Ostlund's *Modern Quantum Chemistry*, followed here for internal consistency across Sections 22–25.

21. The Born–Oppenheimer Approximation and the Molecular Hamiltonian

Non-relativistic molecular quantum mechanics begins from the full electron-plus-nuclei Hamiltonian; every method in this part is, at bottom, a strategy for approximating the eigenstates of its electronic piece once the nuclei are treated as (approximately) fixed external sources.

$$\hat{H} = -\sum_{A} \frac{1}{2M_A}\nabla_A^2 \;-\; \sum_{i} \frac{1}{2}\nabla_i^2 \;-\; \sum_{i,A} \frac{Z_A}{r_{iA}} \;+\; \sum_{i<j} \frac{1}{r_{ij}} \;+\; \sum_{A<B} \frac{Z_A Z_B}{R_{AB}} \tag{21.1}$$

The full molecular Hamiltonian in atomic units: nuclear kinetic energy (mass M_A, index A), electronic kinetic energy (index i), electron–nucleus attraction, electron–electron repulsion, and nucleus–nucleus repulsion, in that order.

$$\hat{H}_{\text{elec}}\,\psi_{\text{elec}}(\mathbf{r};\mathbf{R}) = E_{\text{elec}}(\mathbf{R})\,\psi_{\text{elec}}(\mathbf{r};\mathbf{R}), \qquad \hat{H}_{\text{elec}} = -\sum_{i} \frac{1}{2}\nabla_i^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i<j} \frac{1}{r_{ij}} \tag{21.2}$$

The Born–Oppenheimer approximation: writing the total wavefunction as the product $\Psi(\mathbf{r},\mathbf{R}) \approx \psi_{\text{elec}}(\mathbf{r};\mathbf{R})\,\chi_{\text{nucl}}(\mathbf{R})$ and exploiting $M_A \gg 1$ decouples Eq. (21.1) into this electronic Schrödinger equation, solved at each *fixed* nuclear configuration $\mathbf{R}$, with the nuclear coordinates entering only as parameters.

$$\left[-\sum_{A} \frac{1}{2M_A}\nabla_A^2 + E_{\text{elec}}(\mathbf{R}) + \sum_{A<B} \frac{Z_A Z_B}{R_{AB}}\right] \chi_{\text{nucl}}(\mathbf{R}) = E\,\chi_{\text{nucl}}(\mathbf{R}) \tag{21.3}$$

The nuclear Schrödinger equation on the resulting potential energy surface (PES) $E_{\text{elec}}(\mathbf{R}) + V_{NN}(\mathbf{R})$; every notion of equilibrium geometry, vibrational frequency, and reaction barrier in this part is a statement about the shape of this surface, obtained by solving the electronic problem at many values of $\mathbf{R}$.

$$\mathbf{d}_{ij}(\mathbf{R}) = \big\langle \psi_i(\mathbf{r};\mathbf{R}) \big| \nabla_{\mathbf{R}} \big| \psi_j(\mathbf{r};\mathbf{R}) \big\rangle \tag{21.4}$$

The non-adiabatic (derivative) coupling vector between electronic states i, j: the term neglected by the Born–Oppenheimer approximation, which couples different electronic surfaces whenever nuclear motion is fast enough (or surfaces close enough in energy, as at a conical intersection) that the approximation of the previous equation itself breaks down – the starting point of photochemistry and non-adiabatic molecular dynamics.

$$|\Psi_{\text{CASSCF}}\rangle = \sum_{I \in \{\text{active}\}} c_I\,|\Phi_I\rangle, \qquad \{c_I\}\ \&\ \{\text{orbitals}\}\ \text{co-optimized} \tag{21.5}$$

The complete active space self-consistent field (CASSCF) ansatz: a Full CI expansion (Section 25) restricted to a chemically motivated "active" subset of orbitals, with *both* the CI coefficients c_I *and* the orbitals themselves optimized simultaneously – the standard multireference starting point whenever a single Hartree–Fock determinant (and hence single-reference MP2/CC) qualitatively fails, e.g. bond breaking, transition metals, and near-degenerate ground states, complementing the DMRG treatment of large active spaces above.

22. Hartree–Fock Theory

Hartree–Fock is the starting point of essentially every wavefunction-based method: it approximates the N-electron wavefunction by a single antisymmetrized product (Slater determinant) of one-electron spin-orbitals, optimized variationally.

$$\Psi_{\mathrm{HF}}(\mathbf{x}_1,\dots,\mathbf{x}_N) = \frac{1}{\sqrt{N!}}\det\big[\chi_1(\mathbf{x}_1)\,\chi_2(\mathbf{x}_2)\cdots\chi_N(\mathbf{x}_N)\big] \tag{22.1}$$

The Slater determinant: the antisymmetrized product of N occupied spin-orbitals, the unique way to build a wavefunction obeying the Pauli exclusion principle from a single set of one-particle states.

$$E_{\mathrm{HF}} = \sum_{i=1}^{N}\langle i|\hat{h}|i\rangle + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\langle ij\|ij\rangle \tag{22.2}$$

The Hartree–Fock energy as the expectation value $\langle\Psi_{\mathrm{HF}}|\hat{H}_{\mathrm{elec}}|\Psi_{\mathrm{HF}}\rangle$: a one-electron (core) contribution $\hat{h} = -\tfrac{1}{2}\nabla^2 - \sum_A Z_A/r_A$ plus a two-electron contribution built from the antisymmetrized integrals of the Preface conventions, which packages the Coulomb repulsion and exchange energy into a single sum.

$$\hat{f}(1) = \hat{h}(1) + \sum_{j=1}^{N}\Big[\hat{J}_j(1) - \hat{K}_j(1)\Big], \qquad \hat{f}(1)\,\chi_i(1) = \varepsilon_i\,\chi_i(1) \tag{22.3}$$

The Fock operator and the canonical Hartree–Fock equations obtained by minimizing Eq. (22.2) subject to orthonormality of the χ_i: an effective one-electron eigenvalue problem in which each electron moves in the mean field of all the others, with $\hat{J}_j$ the Coulomb operator and $\hat{K}_j$ the (non-local) exchange operator defined below.

$$\hat{J}_j(1)\chi_i(1) = \left[\int \chi_j^*(2)\,r_{12}^{-1}\,\chi_j(2)\,d\mathbf{x}_2\right]\chi_i(1), \qquad \hat{K}_j(1)\chi_i(1) = \left[\int \chi_j^*(2)\,r_{12}^{-1}\,\chi_i(2)\,d\mathbf{x}_2\right]\chi_j(1) \tag{22.4}$$

The Coulomb operator $\hat{J}_j$ (the classical electrostatic potential of electron j's charge density) and the exchange operator $\hat{K}_j$ (a purely quantum-mechanical, non-local operator with no classical analogue, arising entirely from antisymmetry); the orbital energies ε_i of Eq. (22.3) are the building blocks of every perturbative correction in Sections 24–25.

$$\hat{H}_{\mathrm{DC}} = c\,\boldsymbol{\alpha}\cdot\hat{\mathbf{p}} + \beta mc^2 - \sum_A \frac{Z_A}{r_A} \tag{22.5}$$

The one-electron Dirac–Coulomb Hamiltonian: the relativistic generalization of the core Hamiltonian $\hat{h}$ underlying Eq. 22.2, with $\boldsymbol{\alpha}, \beta$ the 4×4 Dirac matrices acting on a four-component spinor; four-component (or approximate two-component, e.g. ZORA, X2C) relativistic Hartree–Fock and DFT are required whenever heavy elements make $(Zc^{-1})^2$ corrections non-negligible, well before Z is large enough to require full quantum electrodynamics.

$$V_{\mathrm{ECP}}(r) = \sum_l \sum_k A_{lk}\, r^{n_{lk}}\, e^{-\alpha_{lk} r^2}\, \hat{P}_l \tag{22.6}$$

An effective core potential (ECP / pseudopotential): replaces the chemically inert core electrons of a heavy atom with a fitted, angular-momentum-dependent effective potential acting only on the valence electrons explicitly treated in Eq. 21.1, both reducing the electron count and – in a relativistic ECP – absorbing scalar relativistic effects (which would otherwise require the full Dirac–Coulomb treatment above) into a handful of fitted parameters A_{lk}, α_{lk}.

23. Basis Sets and the Roothaan–Hall Equations

Solving the Hartree–Fock equations in practice requires expanding the unknown molecular orbitals in a finite basis, converting the integro-differential eigenvalue problem of Eq. (22.3) into an ordinary (generalized) matrix eigenvalue problem.

$$\chi_i = \sum_{\mu=1}^{K} C_{\mu i}\, \varphi_\mu \tag{23.1}$$

The LCAO (linear combination of atomic orbitals) expansion: each molecular spin-orbital is expanded in a fixed, finite set of K basis functions φ_μ (typically atom-centered), reducing the search for χ_i to a search for the expansion coefficients $C_{\mu i}$.

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon} \tag{23.2}$$

The Roothaan–Hall equations: the matrix form of Eq. (22.3) in the AO basis, with Fock matrix $F_{\mu\nu} = \langle\varphi_\mu|\hat{f}|\varphi_\nu\rangle$, overlap matrix $S_{\mu\nu} = \langle\varphi_\mu|\varphi_\nu\rangle$ (non-diagonal for a non-orthogonal AO basis), coefficient matrix $\mathbf{C}$, and diagonal orbital-energy matrix $\boldsymbol{\varepsilon}$; because $\mathbf{F}$ itself depends on $\mathbf{C}$ through $\hat{J}, \hat{K}$, this generalized eigenvalue problem is solved iteratively to self-consistency (the "SCF procedure").

$$g(\mathbf{r}; \alpha, l, m, n) = N\, x^l y^m z^n\, e^{-\alpha r^2} \tag{23.3}$$

A (Cartesian) Gaussian-type orbital (GTO), the near-universal choice of basis function φ_μ in molecular quantum chemistry: the Gaussian radial form, unphysical at both $r \to 0$ and $r \to \infty$ compared to the exact hydrogenic (Slater-type) exponential decay, is nonetheless used almost exclusively because products of Gaussians on different centers are themselves Gaussians, making the requisite two-electron integrals analytically tractable.

$$\varphi_{\text{STO-}n\text{G}}(\mathbf{r}) = \sum_{k=1}^{n} d_k \, g_k(\alpha_k, \mathbf{r}) \tag{23.4}$$

A contracted Gaussian-type function fit to a Slater-type orbital (e.g. STO-3G, $n = 3$): a fixed linear combination of n primitive Gaussians g_k, with contraction coefficients d_k and exponents α_k pre-optimized to mimic the correct (Slater) exponential decay while retaining the analytic tractability of individual Gaussian integrals – the standard compromise underlying essentially every modern basis set.

$$E_{\text{BSSE}} = E_{AB}(AB) - \big[E_A(AB) + E_B(AB)\big] \tag{23.5}$$

The counterpoise correction for basis set superposition error: computing each monomer A, B in the *full* dimer basis set (AB) ("ghost orbitals") removes the spurious energy lowering that occurs whenever a finite basis set lets each monomer artificially borrow the other's basis functions – essential for accurate intermolecular interaction energies at any finite-basis correlated level, including the MP2 energy above.

24. Electron Correlation I: Møller–Plesset Perturbation Theory

Hartree–Fock is exact within the space of single Slater determinants but misses *electron correlation* – the instantaneous, beyond-mean-field correlation of electron motion. Møller–Plesset (MP) theory recovers correlation perturbatively, treating the difference between the true and Fock Hamiltonians as a perturbation.

$$\hat{H} = \hat{F} + \hat{\Phi}, \qquad \hat{F} = \sum_i \hat{f}(i), \qquad \hat{\Phi} = \hat{H}_{\text{elec}} - \hat{F} \tag{24.1}$$

The Møller–Plesset partitioning: the zeroth-order Hamiltonian is the sum of Fock operators of Eq. (22.3) (whose eigenstates are the Hartree–Fock determinant and all its excited determinants, with eigenvalue the sum of orbital energies), and the perturbation $\hat{\Phi}$ is the fluctuation potential – the difference between the exact and mean-field electron–electron interaction.

$$E_{\text{MP2}} = \frac{1}{4} \sum_{i,j} \sum_{a,b} \frac{\big|\langle ij \| ab \rangle\big|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} \tag{24.2}$$

The second-order Møller–Plesset (MP2) correlation energy: the leading correction to E_{HF}, expressed entirely in terms of quantities already available from the converged Hartree–Fock calculation (orbital energies and antisymmetrized two-electron integrals over occupied–virtual pairs); it is the cheapest systematically-improvable correlation method and the standard first check on how well Hartree–Fock alone is performing for a given system.

$$E_n^{(2)} = \sum_{m \neq n} \frac{\left|\langle \Phi_m^{(0)} | \hat{\Phi} | \Phi_n^{(0)} \rangle\right|^2}{E_n^{(0)} - E_m^{(0)}} \tag{24.3}$$

The general Rayleigh–Schrödinger second-order energy correction, of which the MP2 formula above is the special case $|\Phi_n^{(0)}\rangle = |\Phi_0\rangle$ (Hartree–Fock reference) and $\hat{\Phi}$ the fluctuation potential: only doubly-excited determinants contribute, by Brillouin's theorem (which kills the singles) together with the Slater–Condon rules (which kill triples and higher for a two-body $\hat{\Phi}$).

$$|\Psi_{\mathrm{F12}}\rangle = |\Psi_{\mathrm{CC}}\rangle + \hat{T}_{\mathrm{F12}}|\Phi_0\rangle, \qquad \hat{T}_{\mathrm{F12}} \propto \sum_{ij,ab} t_{ij}^{ab}\, f(r_{12})\, \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i \tag{24.4}$$

Explicitly correlated (F12) coupled cluster theory: augmenting the excitation operator with a geminal correlation factor $f(r_{12})$ (typically a Slater-type function of the interelectronic distance r_{12} directly) builds in the exact electron–electron cusp condition analytically, dramatically accelerating basis-set convergence relative to conventional CCSD(T) and largely removing the need for the extrapolation for which the counterpoise correction above otherwise compensates only partially.

$$|\Psi_k\rangle = \hat{R}_k\, e^{\hat{T}} |\Phi_0\rangle, \qquad \hat{R}_k = r_0 + \sum_{ia} r_i^a\, \hat{a}_a^\dagger \hat{a}_i + \cdots \tag{24.5}$$

Equation-of-motion coupled cluster (EOM-CC): excited states are reached by acting with a *linear* excitation operator $\hat{R}_k$ on top of the same ground-state $e^{\hat{T}}$ used above, diagonalizing the (non-Hermitian, similarity-transformed) $\overline{H}$ in the space of $\hat{R}_k$; EOM-CCSD is the wavefunction-theory benchmark against which the cheaper TDDFT excitation energies of Section 28 are routinely calibrated.

25. Electron Correlation II: Configuration Interaction and Coupled Cluster Theory

Full electron correlation requires going beyond a single determinant. Configuration interaction (CI) and coupled cluster (CC) theory are the two dominant systematic routes, related by, respectively, a linear and an exponential parametrization of the exact wavefunction in terms of excitations out of the Hartree–Fock reference.

$$|\Psi_{\mathrm{CI}}\rangle = c_0|\Phi_0\rangle + \sum_{i,a} c_i^a |\Phi_i^a\rangle + \sum_{\substack{i<j \\ a<b}} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \cdots \tag{25.1}$$

The configuration interaction wavefunction: a linear combination of the Hartree–Fock reference $|\Phi_0\rangle$ with all singly-, doubly-, … excited determinants; truncating and applying the variational principle (minimizing $\langle\Psi_{\mathrm{CI}}|\hat{H}|\Psi_{\mathrm{CI}}\rangle$ over the c's) gives CIS, CISD, etc., with Full CI (all excitations, in a given basis) exact within that basis.

$$|\Psi_{\mathrm{CC}}\rangle = e^{\hat{T}}|\Phi_0\rangle, \qquad \hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots, \qquad \hat{T}_2 = \frac{1}{4}\sum_{i,j,a,b} t_{ij}^{ab}\, \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i \tag{25.2}$$

The coupled cluster wavefunction: an *exponential*, rather than linear, parametrization by excitation operators $\hat{T}$ (here shown in second-quantized form for the doubles operator $\hat{T}_2$); truncating $\hat{T}$ at doubles (CCSD) or doubles-plus-perturbative-triples (CCSD(T)) already recovers the great majority of the correlation energy for most closed-shell molecules, at far lower cost than the CI truncation of the same excitation rank, because the exponential automatically generates size-consistent disconnected products of the retained excitations (e.g. $\frac{1}{2}\hat{T}_2^2$ contributes connected-looking quadruple excitations).

$$\langle \Phi_0 | \overline{H} | \Phi_0 \rangle = E_{\mathrm{CC}}, \qquad \langle \Phi_i^a | \overline{H} | \Phi_0 \rangle = 0, \qquad \langle \Phi_{ij}^{ab} | \overline{H} | \Phi_0 \rangle = 0, \qquad \overline{H} \equiv e^{-\hat{T}} \hat{H} e^{\hat{T}} \tag{25.3}$$

The coupled cluster energy and amplitude equations: projecting the similarity-transformed Schrödinger equation $\overline{H}|\Phi_0\rangle = E_{\mathrm{CC}}|\Phi_0\rangle$ onto the reference determines E_{CC}, while projecting onto each excited determinant gives a set of coupled, non-linear algebraic equations for the cluster amplitudes $t_i^a, t_{ij}^{ab}, \ldots$ – solved iteratively, since $\overline{H}$ (via the Baker–Campbell–Hausdorff expansion, which truncates exactly at the four-fold commutator for a Hamiltonian with at most two-body interactions) depends on the amplitudes themselves.

$$q_A = Z_A - \sum_{\mu \in A} \sum_{\nu} P_{\mu\nu} S_{\nu\mu}, \qquad P_{\mu\nu} = \sum_{i}^{\mathrm{occ}} C_{\mu i} C_{\nu i}^{*} \tag{25.4}$$

Mulliken population analysis: partitioning the total electron count among atoms using the density matrix $P_{\mu\nu}$ and overlap matrix $S_{\mu\nu}$ of Eq. 23.2 defines an atomic partial charge q_A – a basis-set-dependent (and for diffuse basis sets, notoriously unreliable) but ubiquitous first diagnostic of where electron density has accumulated or been depleted in a molecule.

$$\alpha(\omega) = \sum_{n} \frac{f_n}{\omega_n^2 - \omega^2} \tag{25.5}$$

The dynamic polarizability, summed over the same oscillator strengths f_n and excitation energies ω_n of Eq. 28.3: its poles reproduce the absorption spectrum, and its static limit $\alpha(0)$ enters the induction energy of the SAPT decomposition above – the single response function connecting excited-state electronic structure to a wide range of linear spectroscopic and dielectric observables.

26. Density Functional Theory I: The Hohenberg–Kohn Theorems and Kohn–Sham Equations

Density functional theory replaces the $3N$-dimensional many-electron wavefunction by the three-dimensional electron density $\rho(\mathbf{r})$ as the fundamental variable – in principle exact, and in practice the dominant tool of modern computational chemistry for systems too large for the wavefunction methods of Sections 24–25.

$$E[\rho] = F[\rho] + \int v_{\text{ext}}(\mathbf{r})\,\rho(\mathbf{r})\,d\mathbf{r}, \qquad E[\rho_0] = E_0 \le E[\rho] \;\; \forall\,\rho : \textstyle\int \rho = N \tag{26.1}$$

The two Hohenberg–Kohn theorems: (1) the ground-state density $\rho(\mathbf{r})$ determines the external potential v_{ext} (up to a constant) and hence the full Hamiltonian, so the universal functional $F[\rho]$ (kinetic plus electron–electron energy) suffices in principle to determine every ground-state property; (2) $E[\rho]$ is variational – minimized at the true ground-state density ρ_0 – exactly as the wavefunction energy is variational in Section 22.

$$\left[-\frac{1}{2}\nabla^2 + v_{\text{eff}}(\mathbf{r})\right]\phi_i(\mathbf{r}) = \varepsilon_i\,\phi_i(\mathbf{r}), \qquad \rho(\mathbf{r}) = \sum_{i}^{\text{occ}} |\phi_i(\mathbf{r})|^2 \tag{26.2}$$

The Kohn–Sham equations: Hohenberg–Kohn's exact but unknown $F[\rho]$ is replaced by the kinetic energy of a *fictitious* non-interacting reference system with the same density, reducing the many-body problem to this set of one-electron equations – formally identical in structure to the Hartree–Fock equations of Eq. (22.3), but with $\hat{K}$ replaced by an (in principle exact) local exchange-correlation potential.

$$v_{\text{eff}}(\mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\,d\mathbf{r}' + v_{\text{xc}}(\mathbf{r}), \qquad v_{\text{xc}}(\mathbf{r}) = \frac{\delta E_{\text{xc}}[\rho]}{\delta\rho(\mathbf{r})} \tag{26.3}$$

The Kohn–Sham effective potential: external, Hartree (classical electrostatic), and exchange-correlation contributions. All of the many-body complexity of the original problem – and all of the approximation in practical DFT – is contained in the single unknown functional $E_{\text{xc}}[\rho]$, the subject of Section 27.

$$\rho_2(\mathbf{r}_1,\mathbf{r}_2) = \rho(\mathbf{r}_1)\rho(\mathbf{r}_2) + \rho(\mathbf{r}_1)\,\rho_{\text{xc}}(\mathbf{r}_1,\mathbf{r}_2) \tag{26.4}$$

The exchange-correlation hole ρ_{xc}: the depletion of the pair density ρ_2 relative to the uncorrelated (Hartree) product $\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)$, obeying the sum rule $\int \rho_{\text{xc}}(\mathbf{r}_1,\mathbf{r}_2)\,d\mathbf{r}_2 = -1$ (exactly one electron of density is excluded from around any reference electron); every approximate $E_{\text{xc}}[\rho]$ in this section is, implicitly, an approximation to this hole.

$$E_{\text{total}} = \sum_i E_{\text{atom}}\big(\mathbf{G}_i\big), \qquad \mathbf{G}_i = \big(G_i^1, G_i^2, \dots\big) \tag{26.5}$$

The general form of a machine-learned interatomic potential (e.g. Behler–Parrinello neural network potentials): a sum of atomic energy contributions, each a learned function of a local, symmetry-invariant descriptor vector $\mathbf{G}_i$ (atom-centered symmetry functions, or their modern equivalents) built from the positions of neighboring atoms – trained on a database of electronic structure calculations from Sections 22–27, and capable of reproducing near-DFT accuracy at a small fraction of the cost, enabling molecular dynamics on time and length scales otherwise inaccessible.

27. Density Functional Theory II: Exchange-Correlation Functionals

The exact exchange-correlation functional $E_{\mathrm{xc}}[\rho]$ of Eq. (26.2) is unknown; practical DFT is the art of constructing accurate approximations to it, organized (Perdew's "Jacob's ladder") by which ingredients of ρ each rung uses.

$$E_x^{\mathrm{LDA}}[\rho] = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}\int \rho(\mathbf{r})^{4/3}\,d\mathbf{r} \tag{27.1}$$

The local density approximation (LDA) exchange energy (Dirac/Slater exchange): the exact exchange energy of a uniform electron gas of density ρ, applied locally point-by-point – the first rung of the ladder, exact in the limit of a slowly-varying density and still the leading term of every functional below.

$$E_{\mathrm{xc}}^{\mathrm{GGA}}[\rho] = \int f\big(\rho(\mathbf{r}), \nabla\rho(\mathbf{r})\big)\,d\mathbf{r} \tag{27.2}$$

The generalized gradient approximation (GGA): the second rung, adding explicit dependence on the local density *gradient* $\nabla\rho$ to correct for the density's actual inhomogeneity (representative functionals: Becke 88 exchange, PBE, PW91).

$$E_{\mathrm{xc}}^{\mathrm{B3LYP}} = E_x^{\mathrm{LDA}} + a_0\big(E_x^{\mathrm{HF}} - E_x^{\mathrm{LDA}}\big) + a_x\big(E_x^{\mathrm{GGA}} - E_x^{\mathrm{LDA}}\big) + E_c^{\mathrm{LDA}} + a_c\big(E_c^{\mathrm{GGA}} - E_c^{\mathrm{LDA}}\big) \tag{27.3}$$

The B3LYP hybrid functional: the third rung, mixing in a fraction of exact (Hartree–Fock-type) exchange E_x^{HF}, computed with the Kohn–Sham orbitals of Eq. (26.2), to correct the self-interaction error inherent in LDA/GGA exchange; the coefficients $(a_0, a_x, a_c) = (0.20,\, 0.72,\, 0.81)$ are Becke's original semi-empirical fit to atomization energies, ionization potentials, and proton affinities, not derived from first principles – B3LYP's continued dominance in the applied chemistry literature is as much an empirical accident of that fit as a matter of physical principle.

$$E_{\mathrm{xc}}^{\text{meta-GGA}}[\rho] = \int f\big(\rho(\mathbf{r}), \nabla\rho(\mathbf{r}), \tau(\mathbf{r})\big)\,d\mathbf{r}, \qquad \tau(\mathbf{r}) = \frac{1}{2}\sum_{i}^{\mathrm{occ}} |\nabla\phi_i(\mathbf{r})|^2 \tag{27.4}$$

The meta-GGA rung of Jacob's ladder: a further ingredient, the Kohn–Sham kinetic energy density $\tau(\mathbf{r})$, is added on top of ρ and $\nabla\rho$ (representative functionals: TPSS, SCAN, r^{2}SCAN), improving on GGA accuracy at comparable cost since τ is already available from the occupied Kohn–Sham orbitals.

$$E_{\mathrm{disp}} = -\sum_{A<B}\sum_{n=6,8,\ldots} f_{\mathrm{damp},n}(R_{AB})\,\frac{C_n^{AB}}{R_{AB}^n} \tag{27.5}$$

The Grimme-type (DFT-D3) empirical dispersion correction: standard semi-local functionals (Sections 27's LDA/GGA/hybrid rungs) miss long-range London dispersion entirely, so an atom-pairwise C_n/R^n correction, damped at short range to avoid double-counting, is added on top – now routine in essentially every production DFT calculation on non-covalently bound systems.

28. Time-Dependent DFT and Excited States

Ground-state Kohn–Sham DFT says nothing directly about electronic excited states or optical spectra. Time-dependent DFT extends the Hohenberg–Kohn framework to time-dependent densities and potentials, and its linear-response formulation is the standard route to vertical excitation energies.

$$i\frac{\partial}{\partial t}\phi_i(\mathbf{r},t) = \left[-\frac{1}{2}\nabla^2 + v_{\text{eff}}[\rho](\mathbf{r},t)\right]\phi_i(\mathbf{r},t) \tag{28.1}$$

The time-dependent Kohn–Sham equations, following from the Runge–Gross theorem (the time-dependent analogue of the first Hohenberg–Kohn theorem): the time-dependent density $\rho(\mathbf{r},t)$ determines the time-dependent external potential uniquely (up to a purely time-dependent function), licensing the same non-interacting-reference construction as Eq. (26.2).

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix}\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \omega \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \tag{28.2}$$

The Casida equations: linearizing the time-dependent Kohn–Sham equations about the ground state converts the search for excitation energies ω into this generalized eigenvalue problem over occupied–virtual transition pairs, with $\mathbf{A},\mathbf{B}$ built from orbital energy differences plus Coulomb and exchange-correlation-kernel coupling between transitions; it is, structurally, the DFT analogue of the CI/CC excited-state problem of Section 25, and the standard method by which UV–vis absorption spectra are computed in practice.

$$f_n = \frac{2}{3}\,\omega_n \left|\langle 0|\hat{\mathbf{r}}|n\rangle\right|^2 \tag{28.3}$$

The oscillator strength of the transition to excited state n: built from the excitation energy ω_n and the transition dipole moment obtained from the Casida eigenvectors above, this is the quantity directly proportional to the intensity of a computed UV–vis absorption band, obeying the Thomas–Reiche–Kuhn sum rule $\sum_n f_n = N_{\text{elec}}$.

$$\hat{H}_{\text{SO}} = \sum_i \zeta_i(r_i)\,\hat{\mathbf{l}}_i \cdot \hat{\mathbf{s}}_i \tag{28.4}$$

The (one-electron, Breit–Pauli-level) spin-orbit coupling Hamiltonian: couples orbital and spin angular momentum with atomic-number-dependent strength $\zeta_i \sim Z_i^4$, responsible for fine structure, intersystem crossing (the mechanism behind phosphorescence), and the breakdown of pure singlet/triplet labeling of electronic states – a relativistic correction of the same physical origin as, but distinct in application from, the Dirac–Coulomb Hamiltonian above.

$$E_{\text{vib}} = \sum_i \hbar\omega_i\Big(v_i + \frac{1}{2}\Big) + \sum_{i\le j} x_{ij}\Big(v_i + \frac{1}{2}\Big)\Big(v_j + \frac{1}{2}\Big) \tag{28.5}$$

Second-order vibrational perturbation theory (VPT2): treating the cubic and quartic terms of the true (anharmonic) potential energy surface as a perturbation on top of the harmonic-oscillator result of Eq. 29.1 introduces anharmonicity constants x_{ij}, correcting the harmonic frequencies toward the experimentally observed fundamentals and enabling prediction of overtone and combination bands invisible to the purely harmonic treatment.

29. Molecular Vibrations, Rotations, and Spectroscopy

Every electronic structure calculation of Sections 22–28 returns, at minimum, an energy at a single nuclear geometry. Molecular spectroscopy is what connects the shape of the potential energy surface of Eq. 2.3(Part I is not being cross-referenced here; see Eq. 21.3) to experimentally observable vibrational and rotational transitions.

$$E_v = \hbar\omega\left(v + \frac{1}{2}\right), \qquad v = 0, 1, 2, \ldots \tag{29.1}$$

The harmonic-oscillator vibrational energy levels of a normal mode of angular frequency $\omega = \sqrt{k/\mu}$ (force constant k from the curvature of the PES at its minimum, reduced mass μ): the standard zeroth-order description of molecular vibration, with anharmonic corrections needed for overtone and combination-band intensities.

$$E_J = B\,J(J+1), \qquad B = \frac{\hbar^2}{2I}, \qquad J = 0, 1, 2, \ldots \tag{29.2}$$

The rigid-rotor rotational energy levels of a linear molecule with moment of inertia I; real spectra additionally show centrifugal distortion and (for non-linear tops) splitting governed by the full moment-of-inertia tensor, but B remains the leading, most directly structure-diagnostic spectroscopic constant.

$$|\langle\psi_f|\hat{\boldsymbol{\mu}}|\psi_i\rangle|^2 \neq 0 \tag{29.3}$$

The electric-dipole transition selection rule: a transition between states ψ_i, ψ_f is allowed (to leading order in light–matter coupling) only if the transition dipole matrix element of the dipole operator $\hat{\boldsymbol{\mu}}$ is nonzero – the origin of every vibrational and electronic selection rule (e.g. that a homonuclear diatomic has no pure rotation-vibration spectrum, since $\hat{\boldsymbol{\mu}} = 0$ identically by symmetry).

$$\mathrm{FC}_{v'v''} = \big|\langle\chi_{v'}|\chi_{v''}\rangle\big|^2 \tag{29.4}$$

The Franck–Condon factor: the overlap of the nuclear vibrational wavefunctions of the initial (v'') and final (v') electronic states, entering the transition dipole moment under the Born–Oppenheimer/Condon approximation and controlling the relative intensities of the vibronic progression in an electronic absorption or emission spectrum.

$$A = \log_{10}\left(\frac{I_0}{I}\right) = \varepsilon\, c\, l \tag{29.5}$$

The Beer–Lambert law: absorbance A is linear in concentration c, path length l, and molar absorptivity ε – the latter, at a given wavelength, computable in principle from the transition dipole moment and oscillator strength above, closing the loop from a computed excited state to a measurable absorption spectrum.

$$|\Psi_{\mathrm{MPS}}\rangle = \sum_{\{\sigma\}} \mathrm{Tr}\left[A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_N}\right] |\sigma_1 \sigma_2 \cdots \sigma_N\rangle \tag{29.6}$$

The matrix product state (MPS) ansatz underlying the density matrix renormalization group (DMRG): each physical index σ_k (occupation of orbital k) carries a matrix A^{σ_k}, with the bond dimension between matrices controlling the amount of entanglement – and hence correlation – the ansatz can capture; DMRG is the method of choice for strongly correlated, quasi-one-dimensional active spaces (transition-metal clusters, stretched bonds) where the single-reference coupled cluster hierarchy of Section 25 converges poorly or not at all.

$$E_{\mathrm{int}} = E_{\mathrm{elst}} + E_{\mathrm{exch}} + E_{\mathrm{ind}} + E_{\mathrm{disp}} \tag{29.7}$$

Symmetry-adapted perturbation theory (SAPT): decomposes an intermolecular interaction energy directly into physically interpretable electrostatic, exchange-repulsion, induction (polarization), and dispersion contributions, rather than as a single supermolecular energy difference – a perturbative alternative to the counterpoise-corrected supermolecular approach of Section 24 that additionally explains *why* two molecules interact the way they do, and the natural benchmark for the empirical dispersion correction of Section 27.

30. Statistical Mechanics of Molecular Systems

Connecting an electronic structure calculation (a $T = 0$, single-geometry quantity) to measurable thermodynamic quantities (finite-T, ensemble-averaged) requires the molecular partition function, built from exactly the vibrational and rotational levels of Section 29.

$$q = q_{\mathrm{trans}} \cdot q_{\mathrm{rot}} \cdot q_{\mathrm{vib}} \cdot q_{\mathrm{elec}} \tag{30.1}$$

The factorization of the single-molecule partition function into translational, rotational, vibrational, and electronic factors – exact if these degrees of freedom are uncoupled (the same Born–Oppenheimer-adjacent separability assumed throughout this part), and the standard starting point of every "ideal-gas" thermochemistry calculation.

$$q_{\text{trans}} = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} V, \qquad q_{\text{vib}} = \prod_{\text{modes}} \frac{1}{1 - e^{-\hbar\omega/k_B T}} \tag{30.2}$$

The translational partition function (particle in a box / free-particle limit) and the harmonic-oscillator vibrational partition function (product over the normal modes of Eq. 29.1, energy measured from the vibrational ground state of each mode).

$$A = -k_B T \ln Q, \qquad U = k_B T^2 \left(\frac{\partial \ln Q}{\partial T}\right)_V, \qquad S = \frac{U - A}{T} \tag{30.3}$$

The bridge from the canonical partition function Q (built from q for an ideal gas of indistinguishable molecules, $Q = q^N/N!$) to the Helmholtz free energy, internal energy, and entropy – the standard route by which a computed vibrational frequency set becomes a computed entropy or free energy of reaction.

$$q_{\text{rot}} = \frac{1}{\sigma} \frac{k_B T}{hcB} \tag{30.4}$$

The rotational partition function of a linear molecule in the high-temperature (classical) limit, in terms of the rotational constant B of Eq. 29.2 and the symmetry number σ (the number of indistinguishable orientations reached by rotation, e.g. $\sigma = 2$ for a homonuclear diatomic) – required to avoid overcounting indistinguishable rotational configurations in the total partition function above.

$$\Delta G_{\text{solv}} = -\frac{q^2}{8\pi\varepsilon_0 a}\left(1 - \frac{1}{\varepsilon}\right) \tag{30.5}$$

The Born solvation energy: the electrostatic free energy of transferring a point charge q (effective radius a) from vacuum into a uniform dielectric continuum of permittivity ε – the zeroth-order model underlying every implicit-solvent method (e.g. PCM, COSMO) used to add solvation effects on top of a gas-phase electronic structure calculation without explicit solvent molecules.

$$Z_N = \oint \mathcal{D}\mathbf{R}(\tau)\, e^{-S_E[\mathbf{R}(\tau)]/\hbar}, \qquad S_E = \int_0^{\beta\hbar} d\tau \left[\frac{M\dot{\mathbf{R}}^2}{2} + V(\mathbf{R})\right] \tag{30.6}$$

The path-integral (ring-polymer) representation of nuclear quantum statistical mechanics: each quantum nucleus is mapped, via the same imaginary-time path integral construction as Eq. 1.2, onto a classical ring polymer of $N \to \infty$ beads connected by harmonic springs – the basis of path-integral molecular dynamics (PIMD), which recovers zero-point energy, tunneling, and other nuclear quantum effects invisible to the classical-nuclei Born–Oppenheimer dynamics run on the electronic PES of Eq. 21.3.

$$G_{\text{cond}} = \frac{2e^2}{h}\, T(E_F) \tag{30.7}$$

The Landauer formula for molecular conductance: the low-bias conductance of a single molecule wired between two electrodes is fixed by the quantum of conductance $2e^2/h$ times the electron transmission probability T at the Fermi energy – the basic equation of molecular electronics, and a direct, measurable consequence of treating electron transport as coherent quantum scattering rather than classical (Ohmic) current flow.

31. Chemical Kinetics and Transition State Theory

A computed potential energy surface also determines reaction rates, via the height of the barrier separating reactants from products and the statistical mechanics of the barrier-top "transition state."

$$k = A\, e^{-E_a/RT} \tag{31.1}$$

The (empirical) Arrhenius equation: rate constant k as a function of the activation energy E_a and a pre-exponential ("frequency") factor A; purely phenomenological, it motivates but does not derive the more fundamental transition state theory result below.

$$k = \frac{k_B T}{h}\, e^{-\Delta G^{\ddagger}/RT} \tag{31.2}$$

The Eyring equation of transition state theory: the rate constant is written in terms of the free energy of activation $\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$ between reactants and the transition state (a first-order saddle point of the PES of Eq. 21.3, identifiable by having exactly one imaginary vibrational frequency in the harmonic analysis of Section 29), with $k_B T/h$ a universal attempt frequency common to every reaction.

$$v = \frac{V_{\max}[S]}{K_M + [S]} \tag{31.3}$$

The Michaelis–Menten equation: the steady-state rate law for a one-substrate enzyme-catalyzed reaction proceeding through a bound intermediate, $E + S \rightleftharpoons ES \to E + P$; K_M (substrate concentration at half-maximal rate) and $V_{\max}$ are extracted from the underlying elementary rate constants via the steady-state approximation, used throughout multi-step reaction kinetics generally.

$$E_{\mathrm{ONIOM}} = E(\mathrm{high}, \mathrm{model}) + E(\mathrm{low}, \mathrm{real}) - E(\mathrm{low}, \mathrm{model}) \tag{31.4}$$

The ONIOM (and, in its simplest two-layer QM/MM form, $E = E_{\mathrm{QM}} + E_{\mathrm{MM}} + E_{\mathrm{QM/MM}}$) multi-layer energy extrapolation: a small, chemically active "model" region is treated at a high (e.g. CCSD(T) or DFT) level, embedded in a larger "real" system treated at a cheaper level (semi-empirical or molecular mechanics) – the standard strategy for applying the accurate but expensive methods of Sections 24–28 to enzymes, materials, and other systems far too large to treat in full at that level.

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}, \rho] + \sum_k \left(\hat{L}_k \rho \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \rho\} \right) \tag{31.5}$$

The Lindblad (GKSL) master equation: the most general Markovian, completely positive evolution of an open quantum system's density matrix ρ, with the unitary von Neumann term generalizing the closed-system evolution implicit throughout Sections 22–28, and the jump operators $\hat{L}_k$ encoding decoherence and dissipation from a surrounding bath – the standard framework for molecular quantum dynamics in a solvent or thermal environment, and the same formalism underlying noise modeling on the quantum hardware of Section 33.

32. Electrochemistry

The electrochemical formulas below connect the equilibrium and non-equilibrium thermodynamics of electron-transfer reactions to the same quantum-mechanical machinery (orbital/state energies, Fermi's golden rule) used throughout this part.

$$E = E^\circ - \frac{RT}{nF} \ln Q \tag{32.1}$$

The Nernst equation: the cell potential E as a function of the standard potential E° and the reaction quotient Q, for an n-electron transfer with Faraday constant F – the equilibrium (zero-current) statement of electrochemical thermodynamics.

$$j = j_0 \left[e^{\alpha n F \eta / RT} - e^{-(1-\alpha) n F \eta / RT} \right] \tag{32.2}$$

The Butler–Volmer equation: the current density j as a function of overpotential η (departure from the Nernst equilibrium potential), exchange current density j_0, and the (typically $\approx \frac{1}{2}$) transfer coefficient α – the kinetic, non-equilibrium counterpart to the equilibrium Nernst equation above.

$$k_{\mathrm{ET}} = \frac{2\pi}{\hbar} |H_{AB}|^2 \, \frac{1}{\sqrt{4\pi \lambda k_B T}} \, \exp\left[-\frac{(\Delta G^\circ + \lambda)^2}{4 \lambda k_B T} \right] \tag{32.3}$$

Marcus theory of electron transfer: derived from Fermi's golden rule applied to the electronic coupling H_{AB} between donor and acceptor diabatic states in the presence of a harmonic (linear-response) solvent/nuclear reorganization energy λ; the famous "inverted region" prediction – that k_{ET} eventually *decreases* as the reaction becomes more exergonic, once $-\Delta G^\circ > \lambda$ – follows directly from the Gaussian form above and was confirmed experimentally more than two decades after Marcus's original 1956 theory.

$$i_p = 0.446\, nFAC \sqrt{\frac{nFvD}{RT}} \tag{32.4}$$

The Randles–Ševčík equation: the peak current i_p in a cyclic voltammetry experiment on a reversible, diffusion-controlled redox couple, as a function of scan rate v, electrode area A, bulk concentration C, and diffusion coefficient D; the characteristic $i_p \propto \sqrt{v}$ scaling (rather than $i_p \propto v$) is the standard diagnostic distinguishing diffusion-controlled from adsorption-controlled electrochemical processes.

$$U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle \xrightarrow{\text{QPE}} \varphi \approx \frac{E}{2\pi} \tag{32.5}$$

Quantum phase estimation (QPE): given a circuit implementing $U = e^{iHt}$ (built from the Trotter decomposition above) and an eigenstate $|\psi\rangle$, QPE reads out the phase φ – and hence the energy eigenvalue E – to n bits of precision using $O(2^n)$ applications of U; unlike VQE, QPE requires a fault-tolerant quantum computer, but offers a rigorous, exponential asymptotic speedup over any known classical method for the general electronic structure problem.

$$E(X) = E_{\text{CBS}} + A\,e^{-BX} \tag{32.6}$$

A standard complete-basis-set (CBS) extrapolation formula: since Gaussian basis sets of increasing cardinal number X (e.g. the cc-pVXZ hierarchy) converge only exponentially toward the true, complete-basis result, computing at two or three successive values of X and extrapolating to $X \to \infty$ recovers most of the remaining basis-set-incompleteness error at far lower cost than a single calculation in a converged, very large basis – routine practice at the correlated wavefunction level of Sections 24–25, where basis-set convergence is otherwise slow.

33. Quantum Information and Quantum Computing for Chemistry

The final section returns to the language of Part I: mapping the electronic structure problem onto qubits is the starting point for quantum algorithms that promise, for strongly-correlated systems where the coupled cluster truncations of Section 25 struggle, an exponential advantage over classical methods.

$$\hat{H} = \sum_{p,q} h_{pq}\,\hat{a}_p^\dagger \hat{a}_q + \frac{1}{2}\sum_{p,q,r,s} h_{pqrs}\,\hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_r \hat{a}_s \tag{33.1}$$

The second-quantized electronic Hamiltonian: the same physics as Eq. (21.1)'s electronic piece, rewritten in the occupation-number (Fock-space) language of fermionic creation/annihilation operators $\hat{a}_p^\dagger, \hat{a}_q$, with h_{pq} and h_{pqrs} the one- and two-electron integrals of a chosen molecular-orbital basis – the universal starting point for both classical coupled-cluster theory (Section 25) and quantum simulation.

$$\hat{a}_j^\dagger \longmapsto \left(\bigotimes_{k<j} Z_k\right) \otimes \frac{X_j - iY_j}{2} \tag{33.2}$$

The Jordan–Wigner transformation: a mapping of fermionic creation operators to strings of Pauli operators, with the tensor product of Z's over all lower-indexed qubits (the "Jordan–Wigner string") enforcing the fermionic anticommutation relations on a qubit register that natively has none; converts the Hamiltonian above into a weighted sum of Pauli strings measurable on a quantum computer.

$$E(\boldsymbol{\theta}) = \langle\psi_0|\, U^\dagger(\boldsymbol{\theta})\, \hat{H}\, U(\boldsymbol{\theta})\, |\psi_0\rangle, \qquad E_0 = \min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}) \tag{33.3}$$

The Variational Quantum Eigensolver (VQE) cost function: a parametrized circuit $U(\boldsymbol{\theta})$ prepares a trial state from a reference $|\psi_0\rangle$ on the quantum processor, the energy is measured, and a classical optimizer updates $\boldsymbol{\theta}$ – a hybrid quantum-classical loop built on exactly the same variational principle as Hartree–Fock (Eq. (22.2)) and Full CI, but with the trial-state family no longer restricted to what is classically representable.

$$|\psi_{\mathrm{UCC}}\rangle = e^{\hat{T}(\boldsymbol{\theta}) - \hat{T}^\dagger(\boldsymbol{\theta})}\, |\Phi_0\rangle \tag{33.4}$$

The Unitary Coupled Cluster (UCC) ansatz: the natural quantum-computing counterpart of Section 25's coupled cluster wavefunction, with the non-unitary $e^{\hat{T}}$ replaced by the manifestly unitary $e^{\hat{T}-\hat{T}^\dagger}$ (implementable, in principle, as a quantum circuit, unlike $e^{\hat{T}}$ itself) – the standard VQE ansatz for molecular ground states, closing the loop from Part I's abstract quantum-information formalism to a concrete, actively-pursued quantum-chemical application.

$$e^{-i\hat{H}t} \approx \left(\prod_k e^{-i\hat{h}_k t/r} \right)^r, \qquad \hat{H} = \sum_k \hat{h}_k \tag{33.5}$$

The (first-order) Trotter–Suzuki decomposition: since the individual Pauli-string terms $\hat{h}_k$ of the Jordan–Wigner-transformed Hamiltonian generally do not commute, $e^{-i\hat{H}t} \neq \prod_k e^{-i\hat{h}_k t}$ exactly, but the error vanishes as the number of Trotter steps $r \to \infty$ – the standard method by which time evolution (and hence, via quantum phase estimation, the eigenvalues of $\hat{H}$) is implemented as an actual gate sequence on a quantum computer.

$$E_{\mathrm{QM/MM}} \subset E_{\mathrm{ONIOM}} \quad \text{(two-layer case)} \tag{33.6}$$

Quantum embedding closes the compendium where Part II began: from the full molecular Hamiltonian of Eq. 21.1, through mean-field (Section 22), correlated wavefunction (Sections 24–25) and density-functional (Sections 26–27) treatments of a fixed active region, to the multi-scale embedding above and the qubit encodings of this section – every method in Part II is, in the end, a different strategy for deciding which correlations in Eq. 21.1 are worth the cost of treating exactly.

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$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

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THE UNITY OF NATURE

$$Q = \sum_i g_i e^{-E_i/k_B T}$$

THERMODYNAMICS

$dU = TdS - PdV + \mu dN$

$F = U - TS$

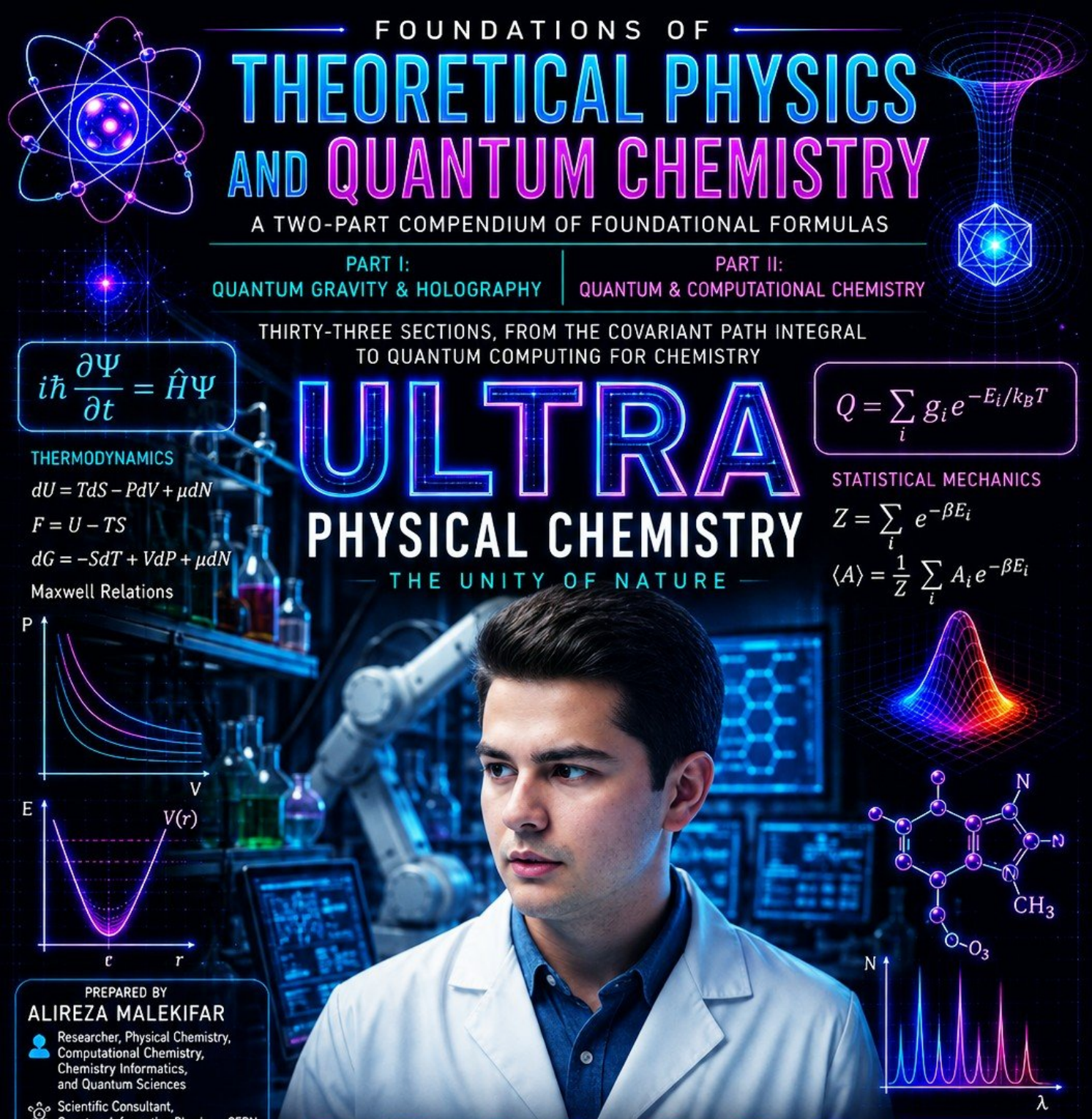
$dG = -SdT + VdP + \mu dN$

Maxwell Relations

STATISTICAL MECHANICS

$$Z = \sum_i e^{-\beta E_i}$$

$$\langle A \rangle = \frac{1}{Z} \sum_i A_i e^{-\beta E_i}$$



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